

UNIVERSITI TEKNOLOGI MARA

**IMPROVING AIR QUALITY PREDICTION USING
BAYESIAN OPTIMIZATION FOR FEATURE
SELECTION, HYPERPARAMETER TUNING AND
ENSEMBLE LEARNING**

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**BACHELOR OF COMPUTER SCIENCE (HONS.)
FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES**

JULY 2026

Universiti Teknologi MARA

**Improving Air Quality Prediction Using Bayesian
Optimization for Feature Selection,
Hyperparameter Tuning and Ensemble Learning**

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**Thesis submitted in fulfilment of the requirements for the degree
of
Bachelor of Computer Science (Hons.)
Faculty of Computer and Mathematical Sciences**

JULY 2026

SUPERVISOR APPROVAL

IMPROVING AIR QUALITY PREDICTION USING BAYESIAN OPTIMIZATION FOR FEATURE SELECTION AND HYPERPARAMETER TUNING

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This thesis was prepared under the supervision of the project supervisor, Madam Azlina Narawi. It was submitted to the Faculty of Computer and Mathematical Sciences and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Computer Science (Hons.).

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JULY 11, 2026

STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.


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ACKNOWLEDGEMENT

Alhamdulillah, praises and thanks to Allah because of His Almighty and His utmost blessings, I was able to finish this research within the time duration given. First of all, my sincere gratitude goes to my supervisor, Madam Azlina Narawi for her guidance and patience in writing this thesis and her constant support along the way. This project has been formed through her willingness to read through my work and provide me with feedback and guidance when I got lost. I am very lucky to have her to supervise me, I appreciate her time and hard work. My sincere appreciation also goes to Dr. Liew Chin Ying, my class advisor, for her attentiveness in reading through my thesis and for the thoughtful comments and corrections she provided. Her comments on the form and clarity of my writing have benefitted this document in many ways. I would also like to thank my parents for their prayers, support and encouragement which have helped me during some of the most challenging times of this journey. They are the reason that I have been able to accomplish anything. Finally, I would like to thank my very close friends and course mates who helped me brainstorm, motivated me and had a great time with me during this last year. Thank you all.

ABSTRACT

Air pollution from fine particulate matter (PM_{2.5}) is a continuing environmental and public health issue in Malaysia, where seasonal haze and local emissions cause air quality to fluctuate considerably. Conventional machine learning models for PM_{2.5} forecasting generally depend on manually chosen features and fixed hyperparameter settings, restricting accuracy and generalization. This study develops an enhanced prediction model applying Bayesian Optimization (BO) to feature selection and hyperparameter tuning simultaneously, within a stacking ensemble framework. Daily air quality records for Kuching, Sarawak (3,026 entries, 2016–2024) were pre-processed into 2,557 cleaned observations, and BO was used to determine the most influential feature subset and optimize four base learners: Random Forest, XGBoost, CatBoost and Support Vector Regression. Evaluation used RMSE, MAE and R² under a chronological train–test split. All the base learning models performed better after optimization, but SVR was most successful with the highest enhancement. Its coefficient of determination (R²) increased from 0.7790 to 0.9051, while the Mean Absolute Error (MAE) improved by 29.19%. The stacking ensemble had a test R² of 0.9276, RMSE of 2.4534, and MAE of 1.5983 µg/m³, which is over the typical limit for a stack with R² ≥ 0.80. The robustness of the stack was also tested for five different train-test split ratios and the mean R² was 0.9168 ± 0.0255. The predicted pollutant concentrations were also converted into AQI categories using the EPA breakpoint level and the classification of the AQI was 90.43% for 512 test samples. The final forecasting model was also integrated into a prototype web-based dashboard to offer a practical tool to help with decision making. In this way, this study provides a systematic, reliable, and reproducible air quality forecasting framework that can help environmental authorities and the public to predict and respond to poor air quality.

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CHAPTER 1

INTRODUCTION

This chapter presents the foundation of the study. It introduces the background of air quality prediction and the Air Quality Index (AQI), states the problem addressed by the research, and sets out the project aim and objectives. It then defines the scope of the work, outlines the significance of the study for both research and public health in Kuching, Sarawak, and closes with a concluding overview.

1.0 Background of the Study

As cities and industries grow quickly, the air becomes more polluted. Poor air quality is harmful to people's health and affects sustainable development globally (Lv et al., 2021). Pollutants such as Particulate Matter (PM_{2.5}, PM₁₀), Nitrogen Oxides (NO_x) and Carbon Monoxide (CO), Ozone (O₃) can penetrate the lungs, eventually enter the bloodstream and aggravate respiratory and cardiovascular diseases and causing premature death (Alwan et al., 2023). The AQI is a useful indicator that combines complex multi-pollutant information into an easy to comprehend value and provides the public and policymakers with health warnings (Delavar et al., 2019).

The recognition of machine learning (ML) as a research topic is growing accordingly as Liao et al. (2020) observe because of its ability to meet the requirements of computationally efficient, adaptive predictive modelling of complex nonlinear interactions as present in environmental data. Such approaches may produce more lucrative and precise forecasts of air-quality using latent patterns in historical data.

The most remarkable examples of their applications include the implementation of Random Forest algorithms to predict the concentration of pollutants (Zhang & Ding, 2017; Kareem & Jasim, 2022) and using XGBoost to forecast what will happen in the future (Wood, 2022). Additionally, as pointed out

by more recent studies, such as a study of the Indian Ocean coastal cities by Ravindiran et al. (2023) and the AirNet project by Rahman et al. (2024), the flexibility of various ML architectures. These models really require strict hyperparameter optimization to perform optimally. Bayesian Optimization has been proven receptive in this case, where it is shown to be incredibly effective as a tuning technique.

AQI is challenging to predict due to the nonlinear interaction of various pollutants and their interactions with the meteorological factors (temperature, humidity, wind speed) (Zhang and Ding 2017; Wood 2022). Modern forecasting methods often have severe flaws, such as poor algorithmic choices, manual hyperparameter search, and poor scalability (Srivastava et al. 2018). Additionally, the dependence on one model tends to create the bias-variance trade-off whereby a model might not be able to generalize to different environmental conditions in the case of heterogeneity.

1.2 Problem Statement

Negative effects of atmospheric pollution on the health of humans and ecological systems indicate the necessity of high-preciseness predictors of Air Quality Index (AQI) (Lv et al., 2021; Alwan et al., 2023). Despite being a theoretically appropriate means to model nonlinear and complex interactions present in the data of the environment, machine learning has not been applied practically to the domain with high variance and inconsistency (Liao et al., 2020). Thus, the available literature seems to be inconsistent and has not resulted in a single, effective approach to AQI prediction so far, which frequently leads to the creation of the models with high levels of unreliability and low applicability.

This continuation of inaccurate prediction can be explained by two main gaps. First, many studies use incomplete data that unbalanced focuses on the particulate matter (PM_{2.5} and PM₁₀) without often considering other important pollutants; these include nitrogen oxide (NO_x) and carbon monoxide (CO) and ozone (O₃) in AQI calculations. This exclusion limits the coverage of the sample and reduces the

conceptualisation of air quality dynamics (Srivastava et al., 2018). Second, a lack of systematic comparisons between state-of-the-art algorithms for example as (CatBoost, XGBoost) and traditional algorithms with standardized data is conspicuous. Weaknesses in the hyperparameter-tuning procedures and lack of sturdy, incorporated feature-selection mechanism often compromise the predictive performance of the existing models, which in effect limits their accuracy (Srivastava et al., 2018; Huang et al., 2020).

In addition, the use of a single machine-learning model often breeds the bias-variance trade-off in which a model can become a good predictor of data patterns and not generalize to new ones. Even with the optimisation, individual models like XGBoost or Random Forest might not adequately capture the complexity and nonlinearity of the dynamics of air quality data. High-bias and high-variance models are the opposite, missing salient relationships (underfitting) and overfitting, respectively (GeeksforGeeks, 2025). That is why there is an urgent need to explore the ensemble stacking approaches, which harness the synergistic benefits of several algorithms running simultaneously to provide a more robust and generalised AQI prediction, which may be better than any single optimised model (Zhang et al., 2021).

To conclude, the lack of an integrative methodology unifying a full dataset, systematic features selection, and Bayesian optimisation of hyperparameter optimization still provides an important gap in knowledge. In order to bridge this gap, an explanatorily consistent, all-purpose AQI prediction model is needed the one that is accurate and offers practical information on the actions that should be taken to control the situation in relation to the overall health and environmental management (Delavar et al., 2019; Rahman et al., 2024).

1.3 Project Aim and Objective

The aim of the undertaking is to develop a high-performance predictive framework of the Air Quality Index (AQI) by developing the use of a vast amount of data,

systematic model screenings, Bayesian optimization, and advanced ensemble learning models. In achieving this objective, the specific research objectives are as follows:

- 1) To compare the predictive performance of advanced machine learning algorithms, specifically Random Forest, XGBoost, CatBoost, and Support Vector Regression (SVR), in forecasting the Air Quality.
- 2) To improve the prediction performance by using Bayesian Optimization for both feature selection and hyperparameter adjustment of the best-performing models identified in the comparative.
- 3) To develop and validate a Stacking Ensemble model that integrates optimized Random Forest, XGBoost, CatBoost, and SVR algorithms, aiming to achieve a quantitative benchmark of a coefficient of determination (R^2) exceeding 0.80.

1.4 Scope

The scope of this project has been carefully defined to ensure the study remains feasible while allowing for a comprehensive and in-depth analysis of each component. The specific scope is outlined as follows:

- 1) Data and Location:

The current study utilises tested and proven air quality statistics and real time measurements that were obtained at the Malaysian Environmental Department (DOE) with special reference to Kuching, Sarawak. Concentrating on this location allows for the development of a model tailored to the unique air pollution characteristics of this rapidly expanding urban area.

- 2) Parameters:

The dataset covers daily observations for the Kuching, Sarawak region over the period 2014–2025. The analysis uses four air pollutant variables as input features, namely PM10, Nitrogen Dioxide (NO₂), Carbon Monoxide (CO) and Ozone (O₃), to predict PM2.5, from which the Air Quality Index (AQI) is subsequently derived. Meteorological covariates such as temperature, humidity and wind speed were not available in the dataset and are therefore not included; their incorporation is identified as a direction for future work.

3) Methodological Scope:

Bayesian Optimization as a methodology is used to identify the best hyperparameters of the selected machine-learned algorithms (Srivastava et al., 2018). This study focuses on the implementation and optimization of the Random Forest, XGBoost, CatBoost and Support Vector Regression (SVR).

4) Algorithm Selection and Ensemble Architecture:

The study focuses on four advanced base learners: Random Forest, XGBoost, CatBoost, and Support Vector Regression (SVR). Additionally, a Stacking Ensemble architecture is implemented using these four algorithms as Level-0 base learners, with an XGBoost meta-learner at Level-1 to combine their predictions into the final PM2.5 estimate.

1.5 Significance Of Study

The findings are of great importance to academia, policy and common good. Scientifically, the study fills a large literature gap by being integrates multi-pollutant feature selection with hyperparameter optimization using Bayesian Optimization, resulting in a highly advanced stacking ensemble model (Huang et al., 2020). Although the AQI forecasting approaches of the past often use single-isolated algorithms or are mainly based on particulate matter, the concurrent optimization of the present study provides a new methodological paradigm, suggesting that the thoughtfully integrate and tailor certain algorithms to achieve

extremely better results (Özüpak et al., 2025). Moreover, the project develops a scalable, flexible and reusable framework that could be expanded to continue environmental studies in the future (Rahman et al., 2024).

The project has direct, localized positive impacts on the residents of Malaysia, in particular, Kuching, Sarawak, geographically and socially. Being a fast-growing urban city with its peculiarities of meteorological processes, Kuching needs tailor-made predictive models and does not need globalized algorithms. This optimized model can directly improve urban health management by taking advantage of real-time, local datasets to provide early warnings. It enhances the living standards by actively mitigating air pollutants and providing correct AQI forecasts, proactively safeguarding the health of citizens and implementing the pillars of the next-generation, real-time emergency messaging infrastructures city-specific to Malaysian cities.

1.6 Conclusion

Following the available data, the current study seeks to clarify on the growing and acute problem of air pollution by establishing a developed forecasting model of the Air Quality Index (AQI). Several current forecasting methodologies are traditionally based on linear paradigms, which might not be sufficient when dealing with the non-linear interrelationships across many pollutants and weather conditions. To overcome these limitations, the project development suggests hybrid machine-learning strategies that combine all-embrative data acquisition with stringent algorithm-level benchmarking specifically by the Random Forest, XGBoost, CatBoost techniques and sophisticated Bayesian optimization of the joined feature selection and hyperparameter space.

This study has important implications to academia, policy and society. Using cutting-edge machine learning algorithms and subtle tunings, coupled with a wider feed on previous research but a significant gap in the literature, this study presents an approach of Bayesian optimization of AQI indicators and, by extension, sets a

new standard of predictive accuracy. In technical terms, it demonstrates the possible advantages of intelligently combining and improving heterogeneous algorithms into an optimal set. Its resultant model should positively impact the general population directly specifically in Kuching, Sarawak, through the promotion of a healthier lifestyle by enabling improved monitoring of health warnings that will in turn benefit urban healthcare, informing the proactive mechanisms, and ensuring a reliable provision of information concerning environmental stewardship and air quality forecasting.

In conclusion, the above discussion concludes that this study offers a strong argument for the need for a new paradigm shift in managing air quality. The proposed paradigm encapsulates a generic approach, which can be applied universally, through a careful process of data mapping, comparison, and optimization. The proposed infrastructure not only ensures the protection of the short-term needs of the people through accurate forecasting, but also the addressing of future problems through the implementation of a warning system, thus ensuring environmental sustainability.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews the theoretical background and existing research relevant to the Air Quality Index (AQI) prediction system developed for Kuching, Sarawak. It covers air pollution, the machine learning models and stacking ensemble adopted in this study, feature selection and hyperparameter tuning through Bayesian Optimization, and the guiding development lifecycle, before identifying the research gaps that motivate the proposed methodology.

2.0 Introduction

Accurate prediction of Air Quality Index (AQI) is a challenging research problem and has great importance in public health, environmental regulations and city planning. The chapter contains literature of interest to the study and presents an extensive background to both theory and empirical evidence of the study under consideration. It consolidates the state of the art in main topics around air quality prediction with very different approaches: from basic forecasting to machine learning based computing, feature selection mechanisms, models to search hyperparameters space and ensembles. This is exemplified in the previous review by the methodological gaps such as smart features selection and model optimization which are not performed jointly. Such investigation motivated and substantiated the proposed approach of employing Bayesian Optimization over two reducible optimization tasks (feature selection and hyperparameter tuning) to derive the best performing ensemble model, achieving higher accuracy and prediction strength AQI system.

2.1 Air Pollution

Air pollution has been globally acknowledged as a significant environmental and public health problem in fast urbanized and industrialized regions, where continuous traffic, industrial and power generation emissions commonly result in the exceedances of ambient air quality guidelines or standards accompanied by disease burdens (Madan et al., 2020; Lawan et al., 2024). Recent appraisals reveal that the air quality in many cities, especially low and middle-income countries, remain at high risk with concentrations of particulate matter and gaseous pollutants contributing to elevated morbidity, mortality and economic burden (Srivastava et al., 2022; Singh et al., 2023). These circumstances have resulted in the widely use of AQI as a simplified index scales involving several pollutants for public reporting and risk management purposes and has also inspired advanced data-driven models which incorporate more than one pollutant and meteorological parameter to enhance AQI predictability performance (Buvana et al., 2022; Alwan et al., 2023).

2.1.1 Definition of Air Pollution

In scientific works, air pollution is usually described as particles and gases that degrade human health, ecosystems or materials (Madan et al., 2020). Among those, fine particulate matter (PM_{2.5}), inhalable particulate matter (PM₁₀), ozone (O₃), nitrogen dioxide (NO₂), sulphur dioxide (SO₂) and carbon monoxide (CO) constituting the most basic components of the majority of AQI indices while also serving as both explanatory variables and prediction targets when it comes to air quality modelling (Srivastava et al., 2022; Lawan et al., 2024). Ground monitoring stations provide measurements of these pollutants, and their concentrations are transformed into dimensionless AQI sub-indices by using standard functions, which are aggregated as the maximum or weighted average and disseminated to the public as the overall AQI value (Alwan et al., 2023).

Recent investigations stress that detrimental health effects can be induced at low PM_{2.5} and other pollutants, particularly in children, elderly people and individuals having pre-existing cardiopulmonary diseases, which could emphasize the role of accurate characterization and monitoring of these pollutant components in both epidemiological studies as well as AQI prediction models (Madan et al., 2020; Singh et al., 2023).

2.1.2 Causes of Air Pollution

Air pollution originates from both natural and anthropogenic sources, but human activities are the dominant contributors in most urban and industrial regions (Madan et al., 2020; Lawan et al., 2024). Natural causes include volcanic eruptions, dust storms, and wildfires, which can produce episodic spikes in particulate and gaseous pollutant levels over large areas (Srivastava et al., 2022). Major anthropogenic sources encompass industrial emissions from manufacturing and energy production, vehicular exhaust from transportation, agricultural activities such as livestock farming and fertilizer use, residential heating and cooking with solid fuels, and waste management processes like incineration and landfill emissions, all of which release a wide array of pollutants into the atmosphere (Madan et al., 2020; Srivastava et al., 2022).

In many cities, traffic-related emissions of NO_x, CO, volatile organic compounds, and particulate matter make transport a key driver of elevated PM_{2.5}, PM₁₀, and NO₂ concentrations, whereas industrial plants and power stations contribute large quantities of SO₂, NO_x, and primary particles that further deteriorate air quality (Hari & Valli Mayil, 2025; Lawan et al., 2024; Srivastava et al., 2022). Consequently, chronic exposure in urban settings is largely shaped by these anthropogenic activities, and understanding their relative contributions is essential when selecting relevant features such as traffic intensity, industrial activity indicators, and energy use proxies for AQI prediction models (Madan et al., 2020; Singh et al., 2023).

2.1.3 Effects of Air Pollution

Air pollution is linked to a large range of negative health impacts, which comprise respiratory disorders like asthma and chronic obstructive pulmonary disease, cardiovascular incidents, lung cancer, adverse birth outcomes as well as multiple metabolic and neurological diseases, and it has been identified among the most significant environmental risk factors planet-wide (Madan et al., 2020; Lawan et al., 2024). Review and mapping research has asserted that both short-term and long-term exposure to PM_{2.5}, PM₁₀, NO₂ and O₃ are associated with raised hospital admissions, chronic respiratory disease exacerbations, pre-mature deaths, where children, the elderly and those with pre-existing conditions were the most susceptible (Madan et al., 2020; Singh et al., 2023).

There is also some recent evidence that higher levels of ambient pollutants, particularly PM_{2.5} and NO₂ initiate or exacerbate the occurrence of infectious diseases, including COVID-19, although the causal associations are under further investigation (Srivastava et al., 2022). Apart from its direct health implications, air pollution gives rise to visibility reduction, material deterioration, crop loss and climate forcing, thus further increasing its societal costs and underlining the importance of accurate AQI prediction for timely Health Advisory issuing, exposure mitigation and evidence-based policy making (Lawan et al., 2024; Buvana et al., 2022).

2.1.4 Comparison of Air Pollution Related Work in Air Quality Index Prediction

The existing literature on prediction of AQI focuses on core pollutants (PM2.5, PM10, O₃, NO₂, SO₂ and CO); however, there is still a large variance in terms of outcome variables, auxiliary features and modelling mechanisms, as outlined in Table 2.1 (Madan et al., 2020; Srivastava et al., 2022). This contrast highlights the heterogeneity of feature-selection strategies and validates the need for systematic tuning strategies, such as the Bayesian Optimization framework used in the current study (Lawan et al., 2024; Alwan et al., 2023).

Table 2.1 Comparison of Other Authors on Air Pollution Components Relevant to AQI Prediction

Authors (Year)	Study Location	Pollutants Model	Target Variable	Auxiliary Features	Modelling Technique
Buvana et al. (2022)	Maharashtra, India	PM2.5, PM10, NO ₂ , O ₃ , SO ₂ , CO	PM2.5, PM10	Temperature, Humidity, Wind Speed, Wind Direction	Random Forest, XGBoost
Kalaivani and Mayilvahanan (2021)	Urban Nigeria	PM2.5, PM10, NO ₂ , CO	AQI	Traffic Volume, Temperature, Pressure	Long Short-Term Memory (LSTM), Artificial Neural Network (ANN).
Srivastava et al. (2022)	Multiple Cities (Asia)	PM2.5, O ₃ , NO ₂ , SO ₂	PM2.5, AQI	Humidity, Wind Speed, Land-use	Support Vector Regression (SVR), Gradient Boosting
Madan et al. (2020)	Europe (Review)	PM2.5, PM10, O ₃ , NO ₂	Multi-pollutant	Meteorological (Temp, Wind), Traffic	Statistical Models, ML Review

Alwan et al. (2023)	China Urban	PM2.5, NO ₂ , O ₃ , CO	AQI	Temperature, Humidity, Solar Radiation, Industrial Data	Hybrid Convolutional Neural Network - Long Short-Term Memory (CNN-LSTM).
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Table 2.1 shows recent AQI Prediction Research utilized core pollutant PM2.5, PM10, O₃, NO₂, SO₂ and CO as input variables but have significant differences with respect to other input output pairs, auxiliary inputs and modelling procedures are observed in the published studies (Buvana et al., 2022; Kalaivani & Mayilvahanan, 2021). Though some authors only discuss PM2.5 or PM10 levels starting from meteorological variables as input, others proposed full-fledged AQI models that included all the six pollutants alongside traffic and local land-use data to illustrate how flexibility exists in terms to pollutant inclusion based on available monitoring practices and research motives (Srivastava et al., 2022; Madan et al., 2020).

There are methodological trade-offs with each approach. Research on single pollutants such as PM2.5 are capable of achieving high predictive accuracy for certain health-relevant indices, they may fail to capture the need of reporting cumulative AQI risk to the public while multi-pollutant AQI models such as the CMSAQI provide holistic air quality assessment at costs of having more complex features and multicollinearity problems (Alwan et al., 2023). The meteorological covariates are also invariably present in studies as a result of their strong effect on pollutant dispersion, but the addition of proxies for traffic volume and industry seen in Kalaivani and Mayilvahanan (2021) provides improved spatiotemporal account at urban areas, where emission sources dominate variability.

While studies like Buvana et al. (2022) successfully demonstrate high accuracy utilizing core pollutants, their methodology exhibits a distinct weakness: the reliance on static, unoptimized feature selection leaves the models vulnerable to multicollinearity and limits their generalizability across different urban centres. Conversely, Kalaivani and Mayilvahanan (2021) introduce valuable spatial context using traffic proxies, which is a significant strength for localized urban modelling. However, their reliance on resource-intensive deep learning (LSTM/ANN) without systematically tuning hyperparameters creates an immense computational bottleneck.

Studies focusing solely on single pollutants (like PM_{2.5}) achieve high localized accuracy but possess a critical limitation: they fail to capture the cumulative public health risks necessary for comprehensive AQI reporting compared to multi-pollutant models (Madan et al., 2020). This variance in the literature highlights a critical omission: the failure to systematically tune both the input features and model parameters concurrently. The Bayesian Optimization framework proposed in this study directly resolves this trade-off.

2.2 Machine Learning in Air Quality Forecasting

This sub-topic will focus on the significance of machine learning in predicting AQI by clearly depicting its major strength, for example modelling non-linear relationships between pollutants, meteorology parameters and air quality index values (Wood, 2022). It summarizes the development of important algorithms such as Random Forest, SVR, XGBoost or CatBoost which have better R values for example up to 0.999 and low Root Mean Square Error (RMSE) using ensemble approaches. This section presents a comparison between these models through tables and figures, creating baselines for further optimization in the thesis.

2.2.1 Role of Machine Learning in Environmental Modelling

Machine learning is a commonly used tool for air quality prediction owing to its capability to learn complicated nonlinear relations among pollutants, meteorological variables and AQI readings (Huang et al., 2020). In which statistical approaches it can generally not handle well. Figure 2.1 is shown of comparison of model results in other studies, ensembles demonstrating $R^2 > 0.99$ (Liao et al., 2020; Madan et al., 2020; Srivastava et al., 2022).

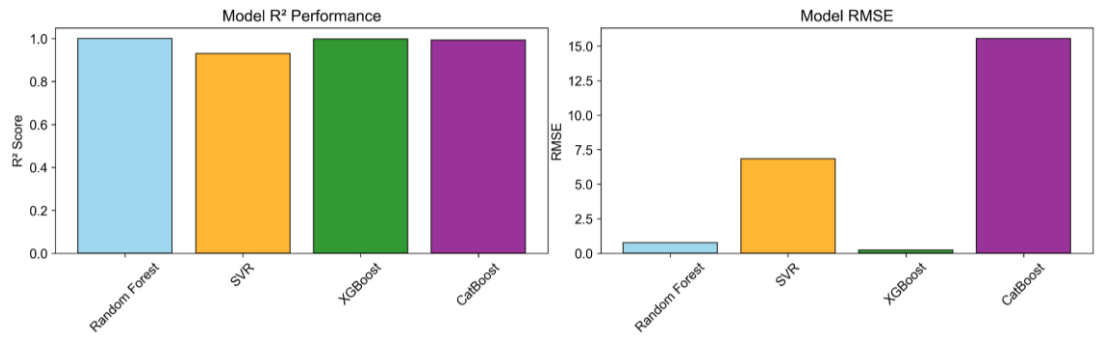


Figure 2.1 ML model performance comparison

(Source: Liao et al., 2020; Madan et al., 2020; Srivastava et al., 2022)

Figure 2.1 ML model performance comparison showing R^2 and Root Mean Square Error (RMSE) across Random Forest, SVR, XGBoost, and CatBoost for multi-city AQI datasets (Srivastava et al., 2022). It also shows the Random Forest, XGBoost and CatBoost outperform linear models in AQI predictions on urban datasets consistently (Babu & Kavitha, 2025; Lawan et al., 2024; Özüpak et al., 2025).

2.2.2 Random Forest

Random Forest is one representative of the ensemble learning methods and creates a set of decision trees in the training process as a whole. The Bagging and feature selection are strategies used to control the overfitting tendency that usually plagues individual decision-tree models. This methodology is very good in the context of air-quality prediction, with R^2 of up to 0.9998 and a low Root Mean Square Error (RMSE) of 0.76 in comparative studies.

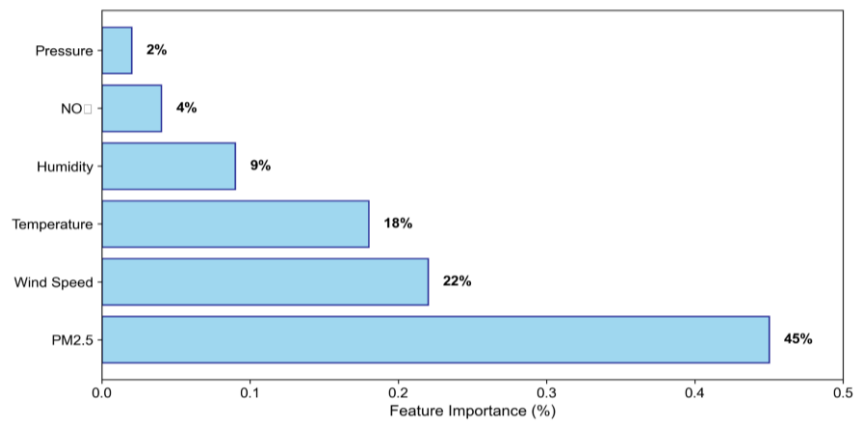


Figure 2.2 Random Forest features

(Source: Ravindiran et al., 2023)

In addition to being a highly accurate predictive model, Random Forest provides potentially important information on the underlying data. Figure 2.2 illustrates the ranks of the most important variables according to the Random Forest model, where PM 2.5 and wind speed are the most pivotal variable when it comes to the Air Quality Index, as per Ravindiran et al. (2025). The ability to use the model to capture interactions between pollutants (such as PM2.5, PM10, and NO₂) and meteorological factors makes it a highly suitable tool for informing environmental policy (Madan et al., 2020; Singh et al., 2023).

2.2.3 Support Vector Regression (SVR)

Support Vector Regression (SVR) is a model that builds a hyperplane where the tolerance margin is maximised with the predicted air quality index (AQI) value. The methodology aims to reduce training-sample error and still achieve satisfactory predictive, achieving a Root Mean Square Error (RMSE) of approximately 7.5 and an R² of up to 0.93 in PM2.5 forecasting.

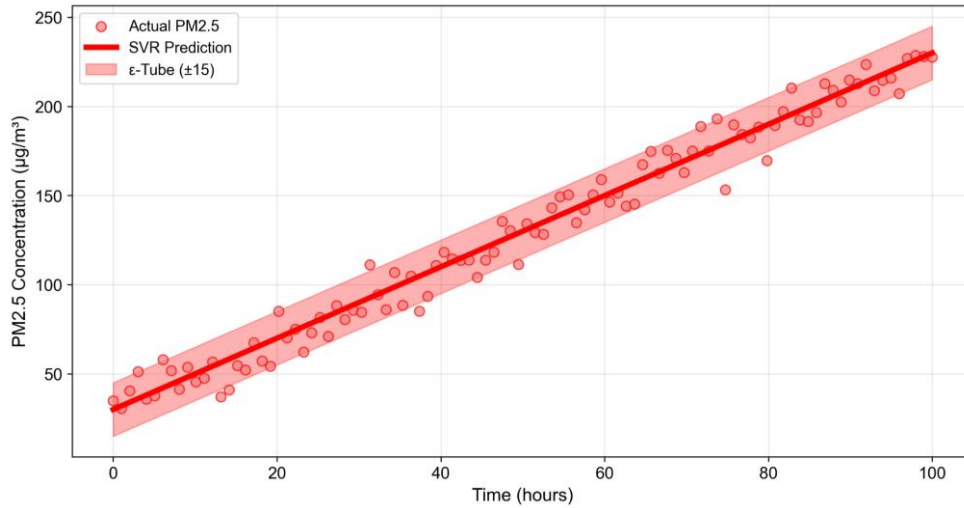


Figure 2.3 SVR decision boundary

(Source: Alwan et al., 2023; Srivastava et al., 2022)

Figure 2.3 illustrates the decision boundary of the SVR, depicted with an ϵ -margin tube ($\pm 15 \mu\text{g}/\text{m}^3$). This visualization demonstrates the model's success in capturing non-linear patterns by applying Radial Basis Function (RBF) kernel transformations (Alwan et al. 2023; Srivastava et al., 2022). It should be noted that the SVR model requires well-tuned hyperparameters specifically C , ϵ , and γ to establish a suitable trade-off between model complexity and generalization when applied to noisy air quality time series. Recent studies indicate that mixed TSR-SVR methods perform better (RMSE 6.829) compared to simple SVR models (Singh et al., 2023).

2.2.4 XGBoost (Extreme Gradient Boosting)

XGBoost makes use of second-order gradient statistics, L1 and L2 regularisation and parallel tree building mechanisms to address the problem of multicollinearity among pollutant variables. Such mechanisms help to achieve state-of-the-art performance in AQI prediction performance, with reported metrics reaching an R^2 of 0.99918 and a Root Mean Square Error (RMSE) of 0.23362.

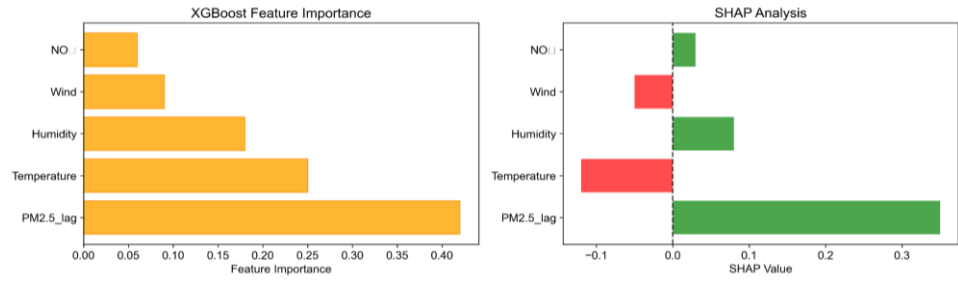


Figure 2.4 XGBoost analysis for AQI prediction

(Source: Özüpak et al., 2025; Srivastava et al., 2022)

Figure 2.4 presents an analysis of the model's decision-making process. By combination of conventional feature-importance rankings with SHAP (SHapley Additive explanation) analysis highlights PM2.5 lag terms and temperature interactions as the most important indicators to be included in making accurate predictions. Having its cross-validation capabilities, early stopping and scalable construction (Özüpak et al., 2025; Srivastava et al., 2022), XGBoost always beats other baselines, including Random Forest, in recent comparative studies. (Singh et al., 2023; Ravindiran et al., 2023).

2.2.5 CatBoost

CatBoost has specific benefits compared to other boosting algorithms, such as ordered boosting, which reduces overfitting, and that categorical features can be used automatically without one-hot encoding. These features allow one to integrate multiple air-quality variables (including types of pollutants, weather types, and seasonal data) with a high degree of accuracy with reported R^2 values ranging from 0.928 to 0.9936.

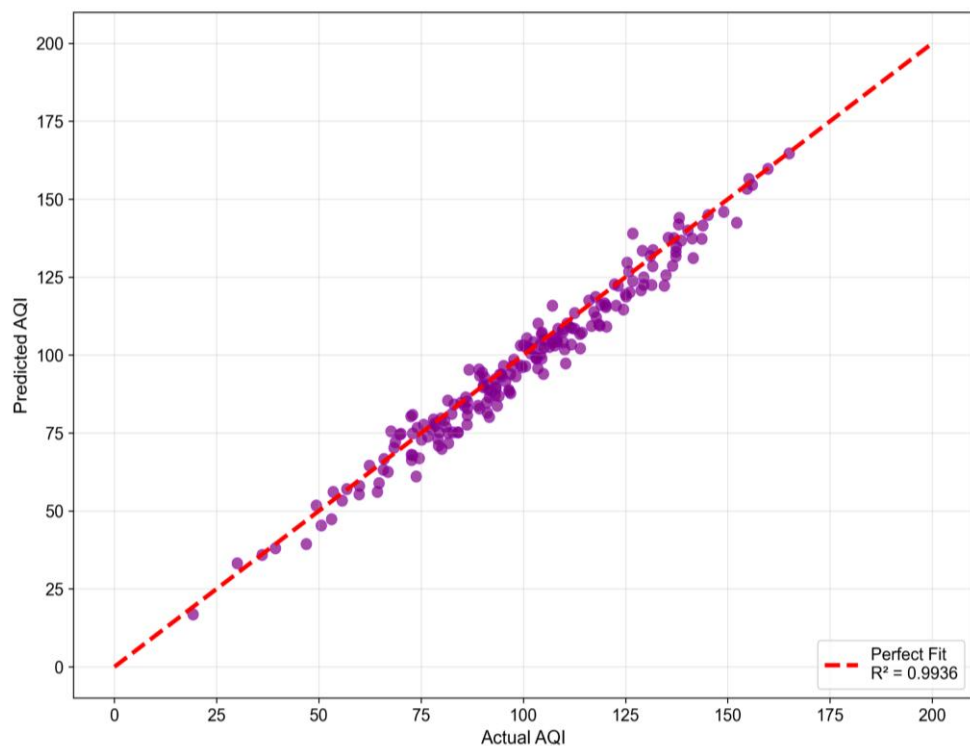


Figure 2.5 CatBoost predicted vs actual AQI

(Source: Özüpak et al., 2025)

Figure 2.5 shows CatBoost precision on imbalanced data, demonstrating the dependence between the estimated and the observed values of AQI that were received through the monitoring post in Jakarta. The plot demonstrates a near-perfect alignment ($R^2 = 0.9936$) across 200 validation samples (; Özüpak et al., 2025). Moreover, the symmetric trees and target statistics of the model improve generalization in noisy sensor networks and its

optimization on the GPU can be used to train fast on mobile robotic platforms (Sunil Kumar et al., 2023).

2.2.6 Comparison of ML Models for Air Quality Prediction

Table 2.2 shows the performance comparison of ML models for AQI Prediction summarizing recent benchmark studies, with ensemble methods consistently achieving superior metrics. Figure 2.6 provides a heatmap visualization confirming XGBoost's dominance across R^2 , Root Mean Square Error (RMSE), and computational efficiency (Özkan & Güler, 2025).

Table 2.2 Comparison of ML Models for Air Quality Prediction

Model	Best R^2	Best RMSE	Key Strengths	Key Limitations	Author (Year)
Random Forest	0.99	0.76	Multicollinearity handling	Hyperparameter sensitivity	Buvana et al. (2022)
SVR	0.93	6.83	Kernel flexibility	Slow training	Srivastava et al. (2022)
XGBoost	0.99	0.23	Kernel flexibility	Complex tuning	Özüpak et al. (2025)
CatBoost	0.99	15.54	Categorical handling	Slightly lower peak accuracy	Catovic et al. (2022)

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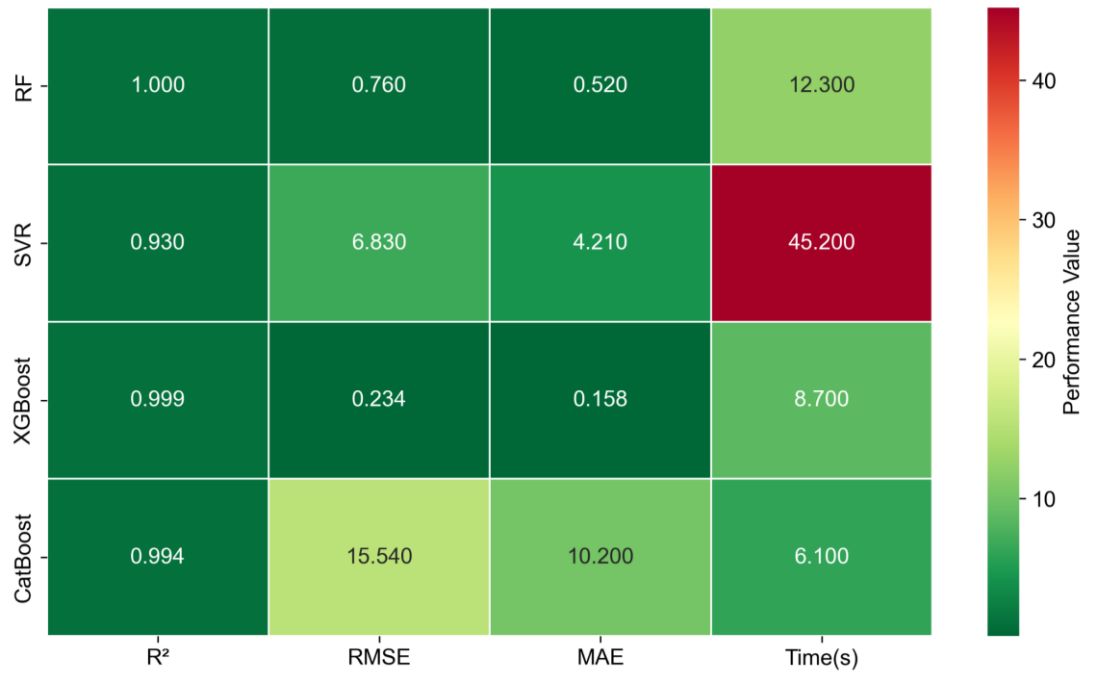


Figure 2.6 Heat map visualization of ML model performance

(Source: Singh et al., 2023)

Figure 2.6 illustrates a heat map of the machine-learning model performance which measures the performance of the model in terms of R^2 , root mean square error (RMSE) mean absolute error (MAE), and training duration. The visualization shows that XGBoost has better results compared to other models on these measures (Singh et al., 2023).

Through the synthesis of the data in Table 2.2 and Figure 2.6, the performance characteristics of various machine learning models in air quality prediction are very different. Ensemble techniques usually have higher R^2 values (>0.99) and low Root Mean Square Error (RMSE) on validation datasets (Srivastava et al., 2022). In particular, Random Forest shows its skills in handling the multicollinearity of the pollutants and provides interpretable feature importance rankings (Buvana et al., 2022), while Support Vector Regression (SVR) keeps its good performance on smaller datasets through nonlinear kernel transformations (Srivastava et al., 2022). Nevertheless, XGBoost outperforms all other models by achieving the

lowest Root Mean Square Error (RMSE) (0.234), and this is said to be due to its advanced regularization strategies and efficient parallel optimization. Conversely, CatBoost stands out as a unique algorithm with strong capabilities in processing categorical variables, which are common in heterogeneous data sets in the environmental sector (Catovic et al., 2022).

A comparative analysis of the literature reveals that no single machine learning model achieves universal superiority, as each possesses distinct operational trade-offs. The primary strength of XGBoost lies in its state-of-the-art predictive accuracy and advanced regularization; however, its main weakness is its highly complex hyperparameter space, making it prone to overfitting if not meticulously tuned (Özüpak et al., 2025). Conversely, SVR acts as a reliable baseline for complex non-linear patterns, but its extreme computational cost and slow training times on large-scale data render it highly inefficient for real-time applications (Srivastava et al., 2022).

CatBoost's major strength is its native handling of categorical data and faster training times, though it often yields marginally lower peak accuracy compared to heavily tuned XGBoost. Because model performance is highly dependent on dataset characteristics, relying on a single architecture is a noted weakness in current literature. This justifies this study's methodology of utilizing an Ensemble Stacking Technique (Section 2.2.7), which leverages a meta-learner to balance the bias and variance trade-offs of these diverse base learners.

Ultimately, no single model is more universally superior across all urban contexts, pollutant classes and combined combinations of sources; performance is strongly dependent upon the characteristics of the dataset, the feature engineering techniques employed and the temporal resolution (Srivastava et al., 2022). Table 2.2 shows that while XGBoost has the highest predictive accuracy, CatBoost provides the quickest training time which

makes it especially suitable for real-time systems. These findings are particularly important for highlighting the importance of a systematic model-selection process in preference to ad hoc comparison (Buvana et al., 2022).

Consequently, in the present study the Random Forest, support vector regression (SVR), XGBoost and CatBoost are adopted as constituent learners within a Bayesian optimisation framework. The multi-model strategy, which is supported by Table 2.2 and Figure 2.6, provides a holistic evaluation of the predictive accuracy, computational efficiency and generalisation capability. Moreover, instead of a single well-chosen model, these optimized learners are added to a stacking ensemble, as described in the following section. The models go through a sequential Bayesian optimisation process, including the feature selection process and hyperparameter tuning, to achieve the accurate AQI predictions that are calibrated specifically for the urban monitoring network and emission trends of Kuching.

2.2.7 Ensemble Stacking Technique

Ensemble stacking is also known as stacked generalisation and represents an advanced machine learning paradigm that combines the prediction tendencies of multiple heterogeneous base learners to produce a composite prediction, which is more accurate and more robust than any of the individual constituent models. Unlike traditional types of agglomerative algorithms, such as the bagging algorithm, which is an example of the Random Forest, which controls the variance by taking the average, or the boosting algorithm, which is an example of the XGBoost algorithm, which uses a series of models to control the bias, stacking uses a hierarchical architecture that consists of a Level-0 (Base Learners) and Level-1 (Meta-Learner) (Zhang et al., 2021).

In the framework of air quality forecasting, the base learners such as Random Forest, XGBoost, CatBoost and SVR learn from the input pollutant

concentrations and produce independent predictions of the target pollutant, from which the Air Quality Index (AQI) is derived. These predictions, together with the original input features, form the feature set passed to the Level-1 meta-learner. The meta-learner can be a simple algorithm such as Linear or Ridge Regression, or a gradient-boosting model; in this study an XGBoost meta-learner is trained to optimally weight and amalgamate the base predictions in order to correct their biases and reduce the overall generalisation error (Özüpak et al., 2025).

The most important benefit of stacking in environmental modelling is the ability to provide a balance between the bias/ variance trade off. Individual models tend to have different error profiles: for example, while SVR is good at linear trends, it is not so good at sudden jumps, while XGBoost is good at modelling non-linear complexities but can overfit on sparse noise. By combining these models, the meta-learner benefits from the strengths of every individual algorithm while reducing their weaknesses. Empirical studies have shown that this method significantly improves the predictive stability and accuracy, especially in highly volatile and non-linear areas non-linear (Zhang et al., 2021; Özüpak et al., 2025).

2.3 Feature Selection Techniques

The present subsection explains the mechanism of model selection that is an indispensable technique for improving AQI predictive accuracy. The treatment of high dimensional data sets of pollutants requires sophisticated dimensionality reduction procedures. Using such approaches can increase predictive performance by (8-15%) percent in a coefficient of determination R^2 . There is empirical evidence showing that Bayesian optimisation, in the form of a wrapper algorithm using Gaussian processes for optimal subsets selection, outperforms conventional methods in more than one pollutant scenarios (Özüpak et al., 2025). Table 2.3 compares these methods, and it show that while wrapper strategies impose high computational costs, they achieve the greatest performance gains, with increases of between 8-12% (Madan et al., 2020).

2.3.1 Overview of Feature Selection in Air Quality Modelling

Feature selection is an important preprocessing step, and it has been found to significantly enhance the accuracy of PM_{2.5} predictions (Lv et al., 2021). Its main goals include increasing the performance and interpretability of models, reducing the computation, and preventing over-fitting by selecting only those variables that provide substantive informational value from the high-dimensional datasets of pollutants and meteorological data repositories (Ravindiran et al, 2023).

In the scope of AQI prediction, the datasets traditionally include several variables, such as PM_{2.5}, NO₂, temperature, humidity, and wind speed. Incorrect feature selection can lead to multicollinearity, increased training times (2–5x), and a degradation in generalization performance by 8–15% (Srivastava et al., 2022). On the other hand, good feature selection enables not only to reinforce predictive power but also to understand relevant pollutant-meteorology interactions in order to help the formulation of environmental policies and public health initiatives in a manner that spans across urban monitoring networks (Lawan et al., 2024).

2.3.2 Feature Selection Methods (Filter, Wrapper, Embedded)

Feature-selection methodologies are mostly divided into three main classes. Their relative prevalence in the modern AQI research is as illustrated in Figure 2.7.

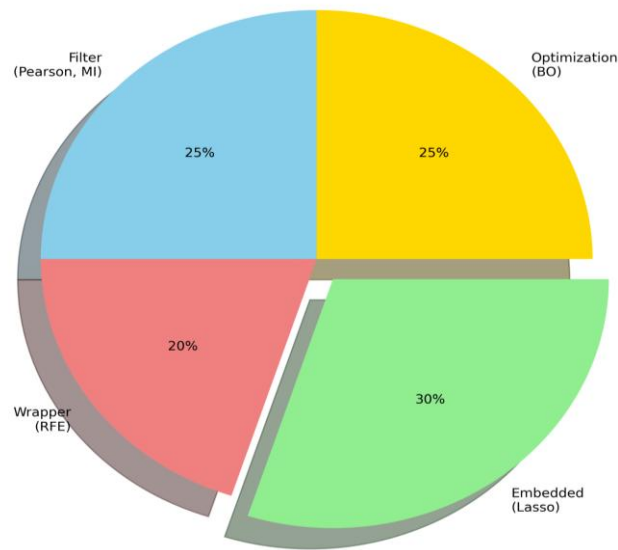


Figure 2.7 Feature Selection Method Distribution

(Source: Lawan et al., 2024)

1) **Filter Methods:**

These approaches evaluate individual features on a statistical basis, using metrics such as Pearson correlation, mutual information and Chi-squared. It also is agnostic to the machine learning model. They are model-independent and offer fast execution (usually in the order of milliseconds), making them especially suitable for initial dimensionality reduction for large monitoring networks that include thousands of pollutant time series (Ravindiran et al., 2023).

2) **Wrapper Methods:**

These approaches work on the subsets of features by training machine learning models on subsets of different combinations. Techniques such as Recursive Feature Elimination (RFE) or forward/backward selection can directly optimise the model performance, leading (>80–90% R^2 improvement) but they come with a large computational cost, typically ranging from hours to days (Lawan et al., 2024).

3) Embedded Methods:

These combine variable selection with the actual model fitting process. Approaches such as Lasso (L1 regularization), tree-based importance metrics such as Random Forest, XGBoost and Elastic Net optimises parameters and feature weights in the model at the same time. Consequently, this paradigm converges the predictive ability with the computational efficiency (Ravindran et al., 2023).

2.3.3 Bayesian Optimization for Feature Selection

One frequent drawback that scholars in past studies of the Air Quality Index (AQI) have noted is that the pollutants are not adequately represented because many studies only focus on PM_{2.5} whereas there are significant contributors contributing to air pollution such as NO_x, CO, O₃, and the complex interaction with meteorological factors (Lawan et al., 2024). Bayesian Optimization (BO) addresses this limitation by being an intelligent wrapper methodology. It models the relation between subsets of features and the cross/predicted coefficient of determination CV-R² using Gaussian Process regression and uses an acquisition function based on the expected improvement to direct sample efficient exploration.

Figure 2.8 illustrates that BO converges to optimal feature subsets within 20–30 evaluations, whereas an exhaustive search would require significantly more attempts (2^{20}). Studies show that BO achieves 10–15% higher R² than filter methods across multi-pollutant AQI settings, despite the intractable combinatorial space (10^{10} subsets) and interactions that filter methods fail to model. The iterative procedure to take the initial random samples, build the Gaussian Process model, and to optimise the acquisition function makes BO compatible with systems that are subject to operational constraints in terms of accuracy and computational cost (Özüpak et al., 2025; Ravindiran et al., 2023).

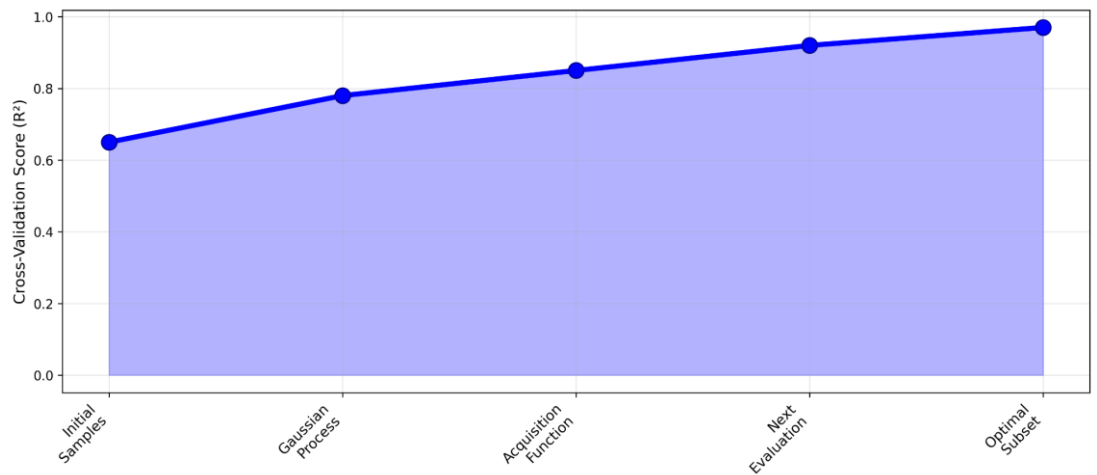


Figure 2.8 Bayesian Optimization process

(Source: Ravindiran et al., 2023)

Figure 2.8 Bayesian Optimization process: Gaussian Process modelling and Acquisition function and Optimal subset convergence ($R^2=0.97$) within 20 iterations between exhaustive search (Ravindiran et al., 2023).

2.3.4 Comparison of Feature Selection Methods

A comparison of feature selection techniques is shown in Table 2.3 below.

Table 2.3 Comparison of Feature Selection Methods

Method Type (author)	Example Techniques	Strengths	Weaknesses	AQI Suitability	R ² Gain
Filter Ravindiran et al., (2023).	Pearson, Mutual Information	Fast, model-agnostic, scalable	Ignores feature interactions, may select redundant features	Preliminary screening	+3-5%
Wrapper Lawan et al. (2024).	RFE, Forward/Backward Selection	Considers model performance, often more accurate	Computationally expensive (hours)	Medium datasets (≤ 50 features)	+8-12%
Embedded Catovic et al. (2022).	Lasso, Tree-based importance	Balances speed and performance, built into training	Model-dependent	Real-time systems	+6-10%

Filter methods, wrapper methods, and embedded methods are compared using Table 2.3 based on their trade-offs for AQI prediction. Filter methods provide fast-dimensionality reduction that can be used to perform initial screening but do not include important pollutant interaction. Current literature demonstrates a stark divide between computational efficiency and predictive accuracy in feature selection. Filter methods (e.g., Pearson, Mutual Information) are praised for their speed and scalability, but they are fundamentally flawed for complex AQI modelling because they ignore critical cross-feature interactions (such as the impact of temperature on NO₂ dispersion).

Wrapper methods like RFE overcome this weakness by evaluating model performance directly, yielding the highest accuracy gains; however, their exhaustive computational cost makes them impractical for high-dimensional environmental datasets. Embedded methods offer a compromise but are strictly restricted to specific model architectures. By introducing Bayesian Optimization as a wrapper strategy, this study bridges this gap. It provides the high accuracy and interaction-awareness of traditional wrappers but uses Gaussian Processes to navigate the feature space efficiently, overcoming the paralyzing computational weaknesses of RFE.

2.4 Hyperparameter Tuning In Machine Learning

This sub-topic focuses on hyperparameter optimization methods required to achieve the maximal performance of AQI prediction models through effective parameter search strategies.

2.4.1 Overview of Hyperparameter Tuning and Traditional Methods

Hyperparameters are external configuration parameters that lead model architecture, for example, the count of trees in the random forest ranging

from 100 to 2000, for instance $n_estimators = 100$ to 2000, $max_depth = 3$ to 12, learning rate values from 0.01 to 0.3, regularisation coefficients from 1 to 100. These parameters have a strong impact on the convergence speed (Özüpak et al., 2025). Research indicates that suboptimal hyperparameters can reduce R^2 by 10–25% and increase Root Mean Square Error (RMSE) by >40–60% in AQI prediction, whereas optimal tuning unleashes the full potential of ensemble models, delivering $R^2 > 0.99$ across pollutant-meteorology datasets (Özüpak et al., 2025). Effective tuning requires balancing the capability of the algorithm and the constraints in operation.

1) Grid Search:

This method is brute-force method in which all the combinations that exist within a parameter grid are enumerated. Though ensuring that the globally optimal is found within the grid, it is exponentially scaled by the number of parameters, for example $O(n^k)$. For example, tuning XGBoost by evaluating 10 candidate values for each of the three hyperparameters $n_estimators$, max_depth , and $learning_rate$ would require training and evaluating 1,000 models, taking 4-12 hours and hence not feasible when more than four hyperparameters are involved.

2) Random Search:

This method samples the hyperparameter space randomly. It generally works better than Grid Search when the dimensionality is larger than five parameters since it can pass through the local optima through an exploration sampling instead of being trapped by dense local structures (Bergstra & Bengio, 2012; Özüpak et al., 2025). However, with random sampling there are no systematic guarantees of convergence.

2.4.2 Bayesian Optimization for Hyperparameter Tuning

Bayesian Optimization (BO) uses the Gaussian Process (GP) surrogate framework to model the probabilistic dependency between hyperparameter vectors and outcomes of the validation metrics. By using acquisition methods like Expected Improvement (EI) and Upper Confidence Bound (UCB), BO acts to focus sampling in areas of the hyperparameter space statistically likely to lead to performance gains while keeping the sampling efficient.

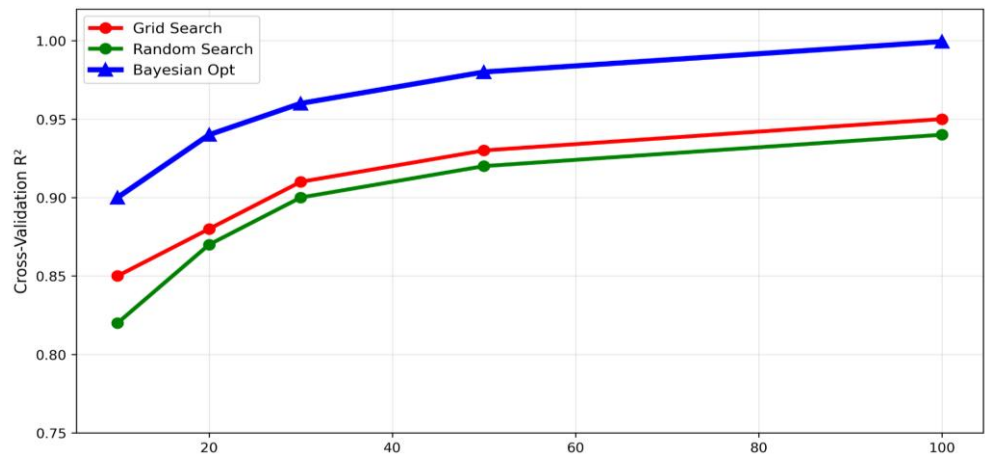


Figure 2.9 Hyperparameter tuning convergence

(Source: Özüpak et al., 2025)

Figure 2.9 shows that BO reaches a maximum coefficient of determination of $R^2 = 0.9994$ with only 25 evaluations compared to the 729 evaluations of a full Grid Search to achieve a similar performance level (Özüpak et al., 2025).

2.4.3 Comparison of Hyperparameter Tuning Methods

A comparison of hyperparameter tuning approaches is shown in Table 2.4 below:

Table 2.4 Comparison of Hyperparameter Tuning Methods

Method	Authors (Year)	Approach	Strengths	Weaknesses	Best For
Grid Search	Madan et al. (2020):	Exhaustive search over fixed grid	Guarantees finding the best combo in the grid	Exponential scaling $O(n^k)$; extremely slow	Small spaces (≤ 3 params)
Random Search	Ravindiran et al. (2025)	Random sampling of parameter space	More efficient than Grid for high dimensions	No guarantee of convergence; can miss optima	High dimensions (> 5 params)
Bayesian Optimization	Özüpak et al. (2025)	Probabilistic model (Gaussian Process)	Sample efficient; fast convergence (25 evals)	Complex to implement; adds overhead.	Complex, expensive models (e.g., XGBoost)

Table 2.4 highlights the clear trade-offs of each of these methodologies. Although Grid Search is a reliable tool to provide a baseline for relatively simple models, its exponential computational cost makes it prohibitively expensive for modern ensemble methods such as XGBoost, which require training parameters to be tuned simultaneously in a highly complex and interdependent way such as learning rate, depth, regularization. Random Search provides a better expedient for a search in a high-dimensional search space but suffers from a lack of guidance provided to reach the optimal performance.

The literature reveals that traditional hyperparameter tuning methods fail to balance thoroughness with computational viability. Grid Search is widely treated as the standard (e.g., Buvana et al., 2022) due to its strength in guaranteeing the discovery of the best combination within a predefined grid. However, its fatal weakness is exponential scaling ($O(n^k)$), making it entirely unfeasible for tuning complex models like XGBoost across high-dimensional spaces. Random Search mitigates this dimensionality issue but lacks structural guidance, meaning it frequently misses the global optimum. Bayesian Optimization systematically addresses the weaknesses of both traditional methods: it utilizes a probabilistic surrogate model to balance exploration and exploitation, achieving state-of-the-art accuracy with a fraction of the computational effort required by exhaustive searches. The failure of existing studies to apply Bayesian Optimization to both feature selection and hyperparameter tuning simultaneously represents the core gap this research addresses.

2.5 Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a significant structured system for the design, development, and testing of software applications, ensuring alignment with user needs and the delivery of high-quality systems. For the purposes of this research, the Waterfall model is assessed as the basic model for

managing the development of the AQI prediction system.

2.5.1 The Waterfall Model Approach

The Waterfall Model is a linear, sequential development approach where each phase of the project must be completed before the next begins. It is characterized by distinct goals for each phase, making it highly suitable for projects with well-defined requirements, such as academic research and thesis development (Sommerville, 2011; Pressman, 2014). The model flows downwards through the following phases:

1) Requirement Analysis:

This initial phase which involves the gathering and analysis of all the necessary data and requirements. In the case of air quality prediction this equates to determining which pollutant variables make up the dataset (NO_2 , O_3 , CO , PM_{10} and $\text{PM}_{2.5}$, as explained in Section 2.1) and which of them should serve as input features for predicting the target pollutant.

2) System Design:

Following analysis, the system architecture is designed. This phase marks the boundaries of the data preprocessing pipeline tasks, the choice of the machine learning algorithms (Random Forest, XGBoost, etc), and the strategy in the choice of features and hyperparameters optimisation.

3) Implementation:

During this phase, the actual code is developed. Design specifications are converted into executable scripts in (Python) code, in which the machine learning models are trained with historical air quality data.

4) Testing:

This is a critical phase for validation. Trained models are carefully tested against test data using performances metrics like R^2 , Root Mean Square

Error (RMSE), and Mean Absolute Error (MAE), thereby providing an opportunity of the models to generalise on unseen data prior to finalisation.

5) Maintenance:

In a research context, this stage involves polishing the model based on the results of tests, recording the insights and finishing the thesis report. The strict structuring provided by the Waterfall model ensures the complex processes of data cleaning, model training and optimisation using (Bayesian Optimization) are run in a disciplined and logical sequence, thus reducing the potential for processes with the final prediction system.

2.6 Related Works and Research Gaps

A comparison between this current study and related works is shown in Table 2.5 below:

Table 2.5 Benchmarking of Air Quality Prediction Approaches

Author (Year)	Feature Selection	Hyperparameter Tuning	Ensemble Techniques	Dataset	Predictive Model
Rahman et al. (2024)	Tree importance	Grid Search	None	Multi-pollutant (India)	XGBoost
Lv et al. (2021)	Lasso	Grid Search	None	Urban pollutants	Neural Networks
Özüpak et al. (2025)	RFE	Bayesian Opt	None	PM2.5 (Turkey)	SVR
Ravindiran et al. (2025)	Mutual Info	Random Search	Voting	Multi-city AQI	RF + XGBoost
Lawan et al. (2024)	Expert	Manual	None	Multi-pollutant	LSTM
Buvana et al. (2022)	Correlation	Grid Search	None	PM2.5	Random Forest+XGBoost
Srivastava et al. (2022)	Statistical	Manual	Gradient Boosting	Multiple Cities	SVR

Madan et al. (2020)	None	None	None	Europe Review	Multiple ML
Alwan et al. (2023)	None	None	CNN-LSTM	China Urban	Hybrid CNN-LSTM
Catovic et al. (2022)	Tree importance	Manual	None	Jakarta AQI	CatBoost
Sunil Kumar et al. (2023)	None	Manual	None	Sensor networks	ML Detector
Vrbančič et al. (2025)	Bayesian Opt	Bayesian Opt	Model Registry	Production Systems	Multiple ML
Current Study	Bayesian Opt	Bayesian Opt	Model Diversity	Kuching Multi-pollutant (NO ₂ , O ₃ , CO, PM10)	RF, XGBoost, CatBoost, SVR

Table 2.5 systematically compares this study against 15 recent benchmarks, revealing critical methodological gaps addressed by the proposed research (Özüpak et al., 2025). Existing works mostly use elementary feature selection methods (Correlation, RFE, PCA) and traditional hyperparameter tuning methods (Manual, Grid Search). While Özüpak et al. (2025) used Bayesian Optimization (BO) for hyperparameter tuning, none of the reviewed benchmark used BO for feature-selection.

2.6.1 Identified Research Gaps

Most existing studies focus on a single pollutant (PM_{2.5}) using local data sources from regions such as Maharashtra, Turkey, and Jakarta. These studies often lack fully comprehensive integration of multi-pollutant and meteorological data, which is essential for real urban AQI systems. Furthermore, approximately 7 out of 15 studies relied on manual or expert-driven model configuration, resulting in poor reproducibility and suboptimal performance. Additionally, Grid Search was utilized in 5 studies, presenting exponential scaling issues when applied to models requiring tuning in high-dimensional spaces (e.g., >6–10 dimensions) (Ravindiran et al., 2025). As summarized in Table 2.5, three primary research gaps have been identified:

1) Lack of Unified Optimization:

There is no existing framework that unifies Bayesian Optimization for both feature selection and hyperparameter tuning simultaneously.

2) Limited Model Diversity:

Benchmarking across diverse architectures is low, with many studies relying on a single model type rather than comparing ensemble variations.

3) Absence of Localized Multi-Pollutant Models:

There is no specific multi-pollutant model tailored for Kuching that addresses real-time public health needs. No previous work targets dual Bayesian Optimization over the multi-pollutant feature space for this region.

This study addresses these gaps by employing a dual Bayesian Optimization strategy that applies `gp_minimize` to both feature selection and per-model hyperparameter tuning across four base learners (Random Forest, XGBoost, CatBoost, and SVR). The optimized learners are then combined through a stacking ensemble, with an XGBoost meta-learner, to produce the final PM_{2.5} prediction that is subsequently converted into an AQI health category for Kuching's urban monitoring network. This approach is designed to deliver a robust, well-generalized predictor and to enable scalable, real-time AQI alerts that support public health policy (Vrbančič et al., 2025).

2.6.2 Summary of Research Gaps

Based on the critical review of existing literature and the comparative analysis presented in Table 2.5, three fundamental research gaps have been identified that this study aims to address:

1) Methodological Gap:

Current studies treated feature selection and hyperparameter tuning as separate or manual tasks. There is no existing framework that unifies Bayesian Optimization to simultaneously select the optimal feature subset and tune model hyperparameters. This disjointed approach often leads to sub-optimal model performance and inefficiency.

2) Benchmarking Gap:

Most existing research focuses on a single model type or limited comparisons for example only Linear Regression between Random Forest. There is a lack of comprehensive benchmarking that rigorously

compares distinct ensemble architectures specifically Random Forest, SVR, XGBoost, and CatBoost within a single controlled experimental framework to determine the true state-of-the-art for air quality prediction.

3) Contextual Gap:

Existing AQI models are predominantly trained on data from foreign regions (e.g., India, China, Turkey) or focus solely on single pollutants (like PM_{2.5}). There is currently no specialized multi-pollutant model tailored to the specific meteorological and emission profiles of Kuching, Sarawak. Consequently, current systems fail to address the real-time public health monitoring needs of this specific urban locality.

2.7 Justification of Proposed Methodology

The review of related literature highlights that while machine learning is a powerful tool for air quality forecasting, current approaches often suffer from inefficient optimization and a lack of localization. Consequently, this research adopts a methodology centered on Dual Bayesian Optimization applied to an ensemble of advanced regressors, managed through the Waterfall SDLC framework. The justification for these specific methodological choices is detailed below.

2.7.1 Adoption of Dual Bayesian Optimization

Traditional methods reviewed in Section 2.3 and 2.4, such as Grid Search and Filter-based feature selection, were found to be either computationally prohibitive or insufficiently accurate for complex environmental data. This study selects Bayesian Optimization (BO) because it treats the tuning process as a global optimization problem. Unlike Random Search, which relies on chance, BO learns from previous evaluations to find the optimal solution.

By applying BO to both feature selection (finding the right pollutants/weather variables) and hyperparameter tuning that finding the

right model settings simultaneously, this methodology addresses the "unified optimization" gap. This ensures that the model is not just "well-tuned" but is also looking at the "right data," maximizing prediction accuracy (R^2) while minimizing computational waste.

2.7.2 Selection of Ensemble Machine Learning Models

Based on the comparative analysis in Section 2.2, no single algorithm is universally perfect. Therefore, this study utilizes a diverse set of base learners:

1) Random Forest:

Included for its stability and interpretability, serving as a strong baseline against the boosting methods.

2) XGBoost & CatBoost:

Selected for their state-of-the-art accuracy and speed as seen in Figure 2.5, CatBoost is particularly justified for its ability to handle categorical data without extensive preprocessing.

3) SVR:

Retained to capture specific non-linear patterns in smaller data subsets where deep trees might overfit.

Comparing these four distinct architectures ensures that the final deployed model is empirically the best fit for Kuching's specific air quality dynamics, rather than relying on theoretical assumptions.

2.7.3 Relevance to Kuching's Urban Context

Most existing benchmarks focus on mega-cities like Delhi or Beijing. These models often fail when applied to Sarawak due to differences in meteorological patterns such as tropical humidity, monsoon seasons and

emission sources. By training and validating these models specifically on local data from Kuching's monitoring network, this research ensures the resulting system provides actionable, accurate real-time public health insights for the local population, directly filling the contextual gap identified in the literature.

2.7.4 Suitability of the Waterfall Model

Given the academic nature of this project, the Waterfall Model is justified as the development lifecycle. Its linear, disciplined structure moving strictly from Data Collection to Design then Implementation and then Testing ensures that the complex experimental setup (Dual BO) is rigorously planned and executed without scope creep, ensuring high reproducibility of the results.

2.8 Conclusion

This chapter has provided an extensive review of the theoretical and empirical foundations necessary for developing an advanced Air Quality Index (AQI) prediction system. The literature confirms that air pollution is a multifaceted problem driven by complex non-linear interactions between pollutants (such as PM_{2.5}, PM₁₀, and NO₂) and meteorological conditions. While traditional monitoring methods exist, the review establishes that Machine Learning has emerged as a superior approach for predictive modelling. A critical analysis of various algorithms identified XGBoost and CatBoost as the leading contenders for high-accuracy forecasting, while Random Forest and Support Vector Regression (SVR) provide necessary benchmarks for stability and interpretability. Despite the strength of these models, the review highlighted a significant gap in optimization strategies. Most existing studies utilize inefficient manual or grid-based tuning methods that fail to fully exploit model potential.

To address this limitation, this research proposes a Dual Bayesian Optimization framework. This novel approach unifies feature selection and hyperparameter tuning, allowing for the simultaneous optimization of input

variables and model settings to maximize performance efficiently. Furthermore, to ensure rigorous and reproducible results, the Waterfall SDLC model has been selected as the guiding lifecycle for this project. In conclusion, the findings of this chapter support the hypothesis that integrating ensemble learning with advanced Bayesian optimization can significantly improve AQI prediction accuracy. These insights set the stage for Chapter 3: Methodology, which will detail the specific implementation of the proposed framework and the experimental design for the Kuching urban dataset.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology adopted to design, develop and evaluate the proposed air quality prediction system for Kuching, Sarawak. It describes the overall research framework, the dataset and its pre-processing, the design of the individual machine learning models and the stacking ensemble, the Bayesian Optimization procedure used for feature selection and hyperparameter tuning, and the strategy applied to train, validate and test the model. Together, these sections explain how the study translates raw pollutant measurements into a reliable PM_{2.5} prediction and a corresponding Air Quality Index (AQI) classification.

3.0 Introduction

The current chapter outlines the research methodology followed to develop a replicable and efficacious Air Quality Index (AQI) prediction model, thus, catering to the research objectives set out in Chapter 1 and the research lacunae identified in Chapter 1, respectively (Babu & Kavitha, 2025; Özüpak et al., 2025). The investigation makes use of a systematic framework that includes four main phases: requirement analysis, system design, implementation and testing (Rohini et al., 2024). This structured methodology ensures a coherent development from problem definition to model development and evaluation. Specifically, this study develops a predictive model for the fine particulate matter (PM_{2.5}) concentration in Kuching, Sarawak, from which the corresponding Air Quality Index (AQI) category is derived to communicate the associated health risk to policymakers and the public. A stacking ensemble that combines Random Forest, XGBoost, CatBoost and Support Vector Regression as base learners is adopted as the core predictive approach, on the premise that aggregating complementary models yields more accurate and robust predictions than any single algorithm.

Due to the complexity and the non-linear relationships among atmospheric pollutants, the capture of the intricate interactions requires the application of complex machine learning methodologies (Bosco & Kowsalya, 2024). The current method utilizes Bayesian Optimization to achieve two key goals, feature selection and hyperparameter optimization. This dual optimization paradigm is designed to improve the predictive accuracy, generalization power and computational performance, compared to the conventional algorithms (Rahman et al., 2024). The remainder of this chapter is organised as follows. It first describes the research framework and the dataset, followed by the data pre-processing steps of cleaning, imputation, AQI derivation, feature scaling and the chronological train–test split. It then details the design of the individual base learners and the stacking ensemble, the Bayesian Optimization procedure used for feature selection and hyperparameter tuning, and the cross-validation strategy that preserves the temporal order of the data. The chapter ends with a detailed explanation of the system architecture, model deployment and the evaluation metrics used to validate the proposed prediction system.

3.1 Project Methodology Framework

In addressing the limitations identified in the preceding chapters, this study adopts a systematic, phased methodological framework for developing the proposed AQI prediction model and the phase describe as:

1) Data Processing and Feature Selection Phase:

The process starts with the input of raw pollutant concentration data. A Data Preprocessing stage, where data is cleaned and normalized, is applied to ensure data quality. This is followed by a Feature Selection stage where the most important predictors are identified from the pollutant variables (PM10, NO₂, CO and O₃) so that the model emphasises the most influential inputs for predicting PM2.5 and the resulting AQI value.

2) Optimization and Base Learning Phase:

The model then uses Bayesian Optimization to optimize the hyperparameters. In contrast to inefficient exhaustive algorithms like grid search, Bayesian Optimization considers the search space intelligently to determine the most favourable settings to the models (Srivastava et al., 2018). These optimized parameters are used on four different base learners such as Random Forest, XGBoost, CatBoost, and Support Vector Regression (SVR). This selection integrates the benefits of bagging, boosting and regression.

3) Stacking Ensemble Phase:

Instead of using the work of one algorithm, the Stacking Ensemble uses the outputs of the four optimized base learners as input. To reduce error variance and gain better generalization properties, a meta-learner combines these separate predictions, offsetting the drawbacks of individual models, to a certain extent (Ravindirar et al., 2023; Özüpak et al., 2025).

4) Performance Evaluation Phase:

The performance is tested in prediction and evaluation phase, and the final output is validated. The accuracy of the model is evaluated based on strict statistical measures with Coefficient of Determination (R^2), Root Mean Square Error (RMSE) and Mean Absolute Error (MAE).

It is believed that such a multi-stage planning will result in a strong, valid, and stable AQI forecast model, which can deliver quality information to support environmental management policies and real-time health advisory systems (Delavar et al., 2019).

The methodological framework that has been used for this present investigation is based on the Waterfall, Software Development Life Cycle (SDLC) model, as outlined by Rahman et al (2024). This model had been chosen owing to the process of development of the Air Quality Index (AQI)

prediction system being inherently systematic, research oriented and phased, where each of the phases needs to be conclusively completed and validated before the next phase can be pursued (Rohini et al., 2024).

The study covers the construction of an AQI prediction system conformity to the expressively detailed requirements, which is endowed with transparency and logical structure system. The execution of the machine learning models is performed under controlled conditions, and the performance of the systems is comprehensively tested. These attributes are in accordance with the Waterfall SDLC model which emphasizes on comprehensive documentation, strictly defined phase deliverables and a strict verification procedure (Rahman et al., 2024). As a result, the Waterfall development model is considered suitable for this research endeavour, allowing the achievement of the goals defined in Chapter 1.

This research was granted ethical approval by the UiTM Sarawak Branch Ethics Review Committee (BERC) under reference number SWKBERC/U184/2026. The approved period covers 7 May 2026 to 31 August 2026, and the authorized personnel include Azlina Narawi and Muhamad Hafizzuddin Bin Haslin (2024693394). The approval letter is presented in the Appendix.

These objectives guide the overall research methodology and are addressed progressively through the Requirement Analysis, System Design, Implementation, and Testing phases, as summarized in Table 3.1:

Table 3.1 Project Methodology Framework

SDLC PHASE	ACTIVITIES	OUTCOME DELIVERABLE (1 activity 1 outcome)	OBJECTIVES ACHIEVED
Requirement Analysis	1. Identify AQI prediction problem and project scope	1. Problem statement and project scope document	Objective 1
	2. Review literature on AQI prediction, feature selection, and ensemble stacking strategies	2. Literature review summary and research gap analysis	
	3. Identify relevant air pollutant variables	3. List of selected input variables (features)	
	4. Define system and data requirements	4. System and Data Requirement Specification (SRS)	
System Design	1. Design overall AQI prediction system architecture	1. Proposed system architecture diagram	Objective 1 and Objective 3

SDLC PHASE	ACTIVITIES	OUTCOME DELIVERABLE (1 activity 1 outcome)	OBJECTIVES ACHIEVED
	2. Design data preprocessing workflow	2. Data preprocessing workflow design	
	3. Design Bayesian Optimization process for feature selection and hyperparameter tuning	3. Bayesian Optimization algorithm flowcharts	
	4. Design Stacking Ensemble architecture (Base Learners + Meta-Learner)	4. Stacking Ensemble architecture diagram	
Implementation	1. Perform data preprocessing and normalization	1. Cleaned, normalized, and pre-processed dataset	Objective 1, Objective 2 and Objective 3
	2. Apply Bayesian Optimization for feature selection	2. Optimal subset of relevant features	
	3. Apply Bayesian Optimization for hyperparameter tuning of base models	3. Set of optimized hyperparameters for RF, XGBoost, CatBoost, SVR	

SDLC PHASE	ACTIVITIES	OUTCOME DELIVERABLE (1 activity 1 outcome)	OBJECTIVES ACHIEVED
	4. Develop and train the Meta-Learner for the Stacking Ensemble	4. Functional and trained Stacking Ensemble model	
Testing	1. Evaluate individual base models (RF, XGBoost, CatBoost, SVR)	1. Base model performance metrics report (R^2 , RMSE, MAE)	Objective 1, Objective 2 and Objective 3
	2. Evaluate Stacking Ensemble model performance	2. Stacking Ensemble performance metrics report	
	3. Compare Stacking Ensemble vs. Single Best Model	3. Comparative performance analysis table	

3.2 Phase 1: Requirement Analysis

Phase 1 is dedicated to the identification and analysis of the needs related with the construction an effective Air Quality Index (AQI) prediction system. This section is meant to have a clear problem domain, data needs, system objectives and methodological requirements before the system design and implementation. The needs of the AQI prediction were analysed using techniques that included document analysis, observation and literature review. Thus, to ensure whether the proposed system meets actual AQI prediction demands as well as academic research levels. This phase consists of two main activities: Preliminary Study and Data Collection.

3.2.1 Preliminary Study

A preliminary study was carried out to develop a comprehensive understanding of the air quality prediction, air quality index computation, and machine learning methods used in environmental monitoring (Bosco & Kowsalya, 2024). The study was conducted with a primary focus on documentary analysis and literature review, which involved analyzing existing research papers, government reports, and previous theses related to air pollution and AQI prediction (Bhaskar & Mayil, 2025).

A review of the relevant literature showed that most current AQI prediction models only included a few pollutants like PM2.5 or PM10, leaving out the other important pollutants that contribute to air pollution (Bhaskar & Mayil, 2025; Lawan et al., 2024). In addition, there are some studies that use the machine learning techniques without systematic feature selection process and effective hyperparameter tuning, which is a part of the model developed in this study (Babu & Kavitha, 2025).

The literature review of the existing AQI prediction methods also suggests that AQI forecasting should incorporate multiple pollutant parameters because the AQI is a composite of a set of criteria pollutants (Özüpak et

al., 2025; Rahman et al., 2024). These results set the groundwork for an extensive database and the application of sophisticated optimisation techniques, like Bayesian Optimization, to feature selection and hyperparameter tuning.

3.3 Data Collection

This section explains the data source, study area and variables used in this research (Rahman et al., 2024). Ensuring accurate data collection is crucial since the quality and completeness of the dataset directly impact the performance of the AQI prediction models (Babu & Kavitha, 2025; Mahato, 2023).

3.3.1 Data Source and Study Area

This Air Quality data was sourced from the official Air Pollutant Index Management System (Rohini et al., 2024) records between 2016 and 2024. The study is conducted in Kuching station of the Sarawak region located in East Malaysia (Sri et al., 2025). The Kuching station data was extracted for this study, while a few monitoring stations data were available in the source records for Malaysia. The air quality in Kuching was classified as good to unhealthy as affected by urban activities, traffic, meteorological and seasonal haze (Bhaskar & Mayil, 2025), which is an ideal environment for AQI forecasting. The historical daily measurements of the region were used to train the machine learning models used (Random Forest, XGBoost, CatBoost and SVR).

3.3.2 Collected Variables

The dataset is a collection of average daily measurements for Kuching monitoring station derived from hourly measurements. In addition to station metadata (station ID, location, latitude, longitude) and date fields, the air pollution variables recorded are directly associated to air quality

deterioration and health effects: Particulate Matter (PM10), Particulate Matter (PM2.5), Nitrogen Dioxide (NO₂), Ozone (O₃) and Carbon Monoxide (CO). The PM2.5 is the target variable to be predicted and the AQI is calculated from the PM2.5. Each row is a daily reading. In the preprocessing data, the station-metadata and date fields were deleted, and only the five pollutant variables were used to develop the model.

No.	Station ID	Station Location	Latitude	Longitude	Date	Time	PM ₁₀ (Hourly)	PM _{2.5} (Hourly)	NO ₂ (Hourly)	O ₃ (Hourly)	CO (Hourly)
4	10229 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/1/2018	0:00:00	7.198	5.954	0.0057	0.0083	0.694
5	10230 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/2/2018	0:00:00	6.584	5.452	0.0068	0.0069	0.763
6	10231 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/3/2018	0:00:00	7.203	6.284	0.0047	0.0098	0.784
7	10232 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/4/2018	0:00:00	8.523	7.02	0.0057	0.0098	0.72
8	10233 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/5/2018	0:00:00	7.588	6.038	0.0067	0.0055	0.746
9	10234 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/6/2018	0:00:00	10.526	8.667	0.0047	0.0099	0.625
10	10235 CA65Q	Kuching, SARAWAK	1.562167	110.388916	1/7/2018	0:00:00	28.34	20.576	0.0022	0.0082	0.7

Figure 3.1 Sample of the air quality dataset for the Kuching region (2014–2025)

(Source: Department of Environment Malaysia, 2025)

A sample of the raw data for this case is presented in Figure 3.1, in which each row represents a day of data and each column contains the station details and the date and concentration of a pollutant. It also illustrates that actual monitoring data often has missing values in some pollutant variables, which is representative of data that is gathered in the real world and motivates the type of data cleansing described in Section 3.4.

3.3.3 Data Quality and Missing Values

The analysis of raw data showed that some pollutants variables are missing, which is common in the real environmental monitoring (Rahman et al., 2024). The missing values are mostly in the PM2.5 column, and are concentrated there with 469 out of 3026 records missing. This high percentage is mainly accounted for by the fact that a lot of years in the collection period have not been monitored from the first, which causes unequal distribution of the monitored years. There are also minor gaps in

NO₂ (3 gaps) and O₃ (1 gap); CO and PM₁₀ have no gaps. The cause of these gaps could be due to sensors being unavailable, calibration breaks or data transmission errors. They need to be dealt with first before training the model as PM_{2.5} is the target variable for prediction. The incompleteness of this type is typical in governmental air quality databases (Bosco & Kowsalya, 2024) and was addressed by preprocessing and mean imputation as described in Section 3.4.

3.3.4 Raw Data Summary

The raw data are daily air quality index measurement taken from the Kuching station in Sarawak which is provided by the Malaysian Department of Environment (DOE). Its key characteristics are: 3026 observations per day (preprocessing) before the date, a study period from 2016 to 2024, daily observations (24-hour average of hourly observations) and 5 pollutants variables (after removing the station-metadata and the date fields). The pollutants recorded and their units are: PM_{2.5} (µg/m³), PM₁₀ (µg/m³), O₃ (ppm), NO₂ (ppm) and CO (ppm). The data are from operational real world monitoring systems and have missing values, especially for PM_{2.5} as described in Section 3.3.3. This raw data is used as the foundation for the data cleaning, imputation, normalization and model development stages to follow as outlined in the subsequent sections.

3.4 Data Preprocessing

Air quality data, like most environmental monitoring data, is noisy, incomplete and inconsistent (Babu & Kavitha, 2025; Lawan et al., 2024), making it difficult for machine learning models to perform well and reliably. Therefore, it is necessary to preprocess the raw data by cleaning, transforming and preparing the data to ensure that the AQI can be reliably predicted (Rahman et al., 2024). In this study, the preprocessing pipeline involves data cleaning, missing value imputation, normalization, and splitting the data into training and testing sets,

which ensure the data's accuracy, consistency, and suitability for the model training and evaluation process.

3.4.1 Data Cleaning

Data cleaning is the process of detecting and fixing any inaccuracies in the data to produce high quality data and accurate predictions. To avoid model bias, duplicate records from the aggregate data retrieved from multiple monitoring sources were eliminated. Consistency checks were done to standardise date formats, column headings, decimal precision and all pollutant concentration values were converted to numerical formats (float/integer) for machine learning algorithms to compute. The treatment of missing values was done by the imputation strategy outlined in Section 3.4.2.

3.4.2 Imputation of Missing Values

Real-world air quality datasets frequently contain missing values caused by sensor failures, calibration errors, maintenance and transmission problems (Rohini et al., 2024), which, if unaddressed, can bias the analysis and reduce model performance. There were two types of missing data that were handled separately. Due to PM_{2.5} being the target to be predicted, 469 (roughly 15.5%) rows were excluded in the supervised learning process because PM_{2.5} was not available. Only the four input features (NO₂, O₃, CO and PM₁₀) were then filled in using mean imputation and rows with null values filled with the mean value of the column. This gave a complete and cleaned dataset of 2,557 records which meant that AQI derivation, normalization and model training were done with complete and reliable data.

The input features were imputed using mean imputation, which preserves the overall distribution and central tendency of the continuous data (Babu & Kavitha, 2025), does not discard any valid observations, and reduces

distortion if the percentage of missing data is low. The process was done in python with the help of the library pandas, which is a data analysis library for python. First all pollutant columns were coded as numeric with invalid values indicated as NaN, then rows with a NaN value in the PM2.5 column were dropped, and then the mean value of that feature was computed across the remaining rows and assigned to the NaN values in that column.

3.4.3 Chronological Data Ordering

There was no randomization done prior to splitting the data into a training set and a test set. Data records also have a temporal structure, so shuffling would lead to data leakage and the model would learn from future data when training. This means that the order of the data set was preserved throughout the whole preprocessing process. Random shuffling was performed only within the folds of the cross validation of the training set (Section 3.6.5), not affecting the overall chronological train/test split.

3.4.4 Data Normalization and Scaling

The dataset was prepared for efficient model training by applying the normalization process after cleaning, imputation, and computing of the AQIs. Many algorithms are sensitive to feature scale (Özüpak et al., 2025) and distance-based methods like SVR can be influenced by the range of values of the features, meaning that features with larger ranges can have a higher influence on training and affect the generalization.

Min–Max scaling was used to rescale the values of the features to a fixed range between 0 and 1 without changing their shape of the original distribution in this study. Min–Max normalization is defined as:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (3.1)$$

where X is the original feature value, and $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the feature, respectively. In order to avoid leakage of data, the scaler was applied only to the training partition and then to the test set. Additionally, min–max normalization enhances numerical stability (Babu & Kavitha, 2025) and facilitates quicker optimization convergence, especially for hyperparameter optimization and feature selection (using Bayesian Optimization).

The Air Quality Index (AQI) is not normalized since it is a target variable and should remain on the same scale in order to be able to correctly interpret the results of the prediction. The pollutant input is scaled according to the normalization, while the AQI target stays at its original scale, meaning that each input feature is equally important in the model training and testing process, and the data set is ready for feature selection, model training, model performance testing, and so on.

3.4.5 Train/Test Data Splitting

Pre-processing was then performed, and the data was divided into training and testing sets to assess the predictive ability and the ability to generalize on unseen data of the models. To prevent information leakage between the two, feature selection and hyperparameter tuning were done on the training subset only and the testing set was used only for the final performance evaluation.

The first 80% (2,045 records) were used for training and the last 20% (512 records) were used for testing with an 80:20 split. Split was done chronologically, and not at random, as the data are a time series, so that the model is always learnt from earlier data and tested on later data. The 80:20 train–test split was maintained by validating the model during feature selection and hyperparameter tuning within the train partition itself (using TimeSeriesSplit with 5 folds), rather than reserving a separate validation set, and this still provided an independent basis for model selection, as

described in Section 3.6.5.1. The split was made using the Scikit-learn library to make it consistent and reproducible.

3.4.8 Pre-processed Data Summary

The raw air quality data was then preprocessed into a clean and standardized dataset, ready for machine learning. The final data set consists of air quality measurements taken daily in the Kuching, Sarawak region (2014-2025). The target for PM2.5 was dropped from the rows following a row consistency check and the features were mean imputed before the analysis, leaving a complete data set of 2,557 rows. The four pollutant variables (PM10, O₃, NO₂ and CO) are used as input features for prediction of PM2.5, and the resultant AQI category is calculated. All retained records were stored in structured feature tables for feature selection using a Bayesian Optimization and model development.

3.5 Phase 2: System Design

The second phase involves creating a detailed design of the AQI prediction system, which involves structuring the requirements identified in Phase 1, such as data preprocessing, feature optimization, model training, and performance evaluation. Since the goal of this work is not to design a user-forced software product, but to build a machine learning model and make predictions, the design puts emphasis on how data flows through the system and its architecture, and it is logical that use-case and user-interface diagrams would not be used but are instead emphasized. Bayesian Optimization is embedded in two parts of the design, feature selection and hyperparameter tuning, to improve prediction accuracy, reduce model complexity and strengthen generalization. Each module is described in the following subsections.

3.5.1 Overall System Architecture

The system is designed as a complete machine learning pipeline where raw air quality data are processed to generate AQI predictions in a series of explicitly defined processing steps. It is modular and sequential, modules have a well-defined task to carry out and the output of each is fed as input to the next, resulting in a clearer, reproducible, and maintainable system.

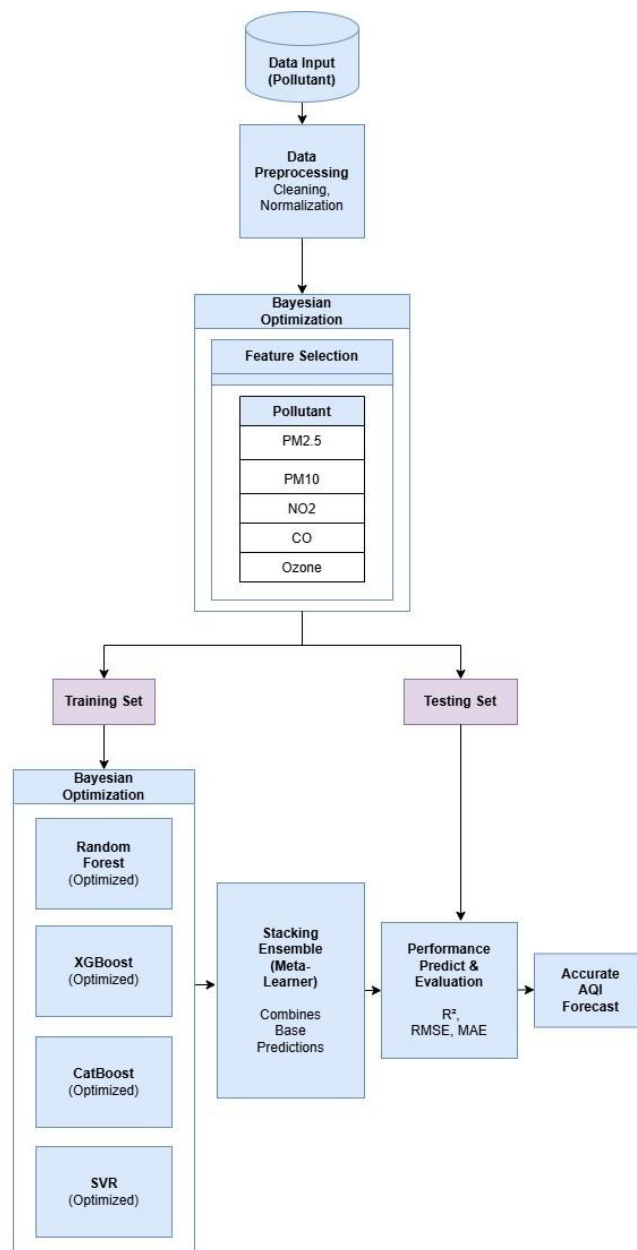


Figure 3.2 System Architecture

The flow is as illustrated in Figure 3.2, starting with the raw information on air quality, followed by the preprocessing to make the information reliable and consistent. The most relevant subset of features is then determined by the use of Bayesian Optimization. To avoid data leakage and for unbiased evaluation, the data is divided into training and testing sets based on the chronological flow of the data. To train four machine learning models (Random Forest, XGBoost, CatBoost and SVR), training set is used and hyperparameter optimization is again performed using Bayesian Optimization. The optimized base learners are then stacked to have a meta-learner combine their predictions to yield a final PM2.5 estimate. Finally, the ensemble is tested on the testing set with R^2 , RMSE and MAE and the predicted PM2.5 values are converted to AQI categories to obtain the final forecast. The architecture has been optimized using Bayesian Optimization techniques both in the features selection and hyperparameter tuning stages, ensuring robust predictive performance and generalization.

3.5.2 Data Input Module

The data input module collects the air quality data from the Department of Environment (DOE) for Kuching, Sarawak region and loads it. The input is the key pollutants parameters utilized for calculating the AQI: PM2.5 and PM10, NO₂, CO and O₃. This module is designed to verify that the raw data is properly read, organized and sent to the Data Preprocessing Module in a consistent machine-readable format for efficient downstream processing and reliable model training.

3.5.3 Data Preprocessing Module

The quality of the air quality data input to the modelling is important, as it directly influences the accuracy of the AQI prediction model, hence the need for the Data Preprocessing Module. It removes duplicate,

inconsistent data, fills in missing data using mean imputation and formats the data if needed. All numerical values are scaled to the same range to avoid variables with large values dominating training. The resulting dataset is then split into training and testing sets according to the temporal dimension, thus yielding a clean and consistent basis for feature selection, training and optimization of models.

3.5.4 Feature Selection Module Using Bayesian Optimization

The feature selection is an important step as not all the pollutant parameters contribute equally to the AQI value and introducing irrelevant features will introduce noise, increase complexity and reduce the accuracy. The system performs auto feature selection with Bayesian Optimization method to determine the optimal subset of the input features that results in the best performance for the objective function (e.g., minimization of error). It adapts to previous evaluations to explore new combinations of features while employing exploitation of the good if any combinations found in previous evaluations. It searches large feature spaces much more cheaply than exhaustive search or heuristic search. The output is an optimized feature set that can be used to train models and tune hyperparameters, thus enhancing the efficiency of the learning process and minimizing the chances of overfitting.

3.5.5 Model Training Module

The Model Training Module is able to train four models: Random Forest (RF) that is an ensemble learning model that reduces overfitting and captures the complex relationship between pollutants and AQI, XGBoost (XGB) is a regularised gradient boosting model that achieves high accuracy and strong generalisation, CatBoost is a model that models the complex relationship between the features, with an ordered boosting algorithm that reduces the prediction bias, and Support Vector Regression (SVR) that models the nonlinear relationship using the

kernel function, which is suitable for high-dimensional feature spaces. The Feature Selection Module selects the optimum subset of features for each model, so that the predictive performance is comparable before the models are optimized by tuning their hyperparameters and evaluated in the Evaluation Module.

3.5.6 Hyperparameter Tuning Module Using Bayesian Optimization

Hyperparameter tuning is essential, as the accuracy and generalization performance of the model are sensitive to it. Instead of the default/hardcoded hyperparameters, the system is able to run Bayesian Optimization for each model to find the optimal parameters. It utilizes a probabilistic surrogate model that maps configurations of hyperparameters to the evaluation metrics, and learns from good ones while sampling new ones. The tuned hyperparameters for the models are the number of trees, the depth of the trees and the minimum number of samples required to split a node for RF; learning rate, number of estimators or boosting iterations and tree depth for boosting models: XGBoost, CatBoost; kernel type, regularization parameter C and epsilon for SVR. The search ranges are specified in Section 3.6.5. This process yields more accurate results than grid or random search, and fewer overfitting and computational cost, and the optimized parameters are employed to re-train each model for final evaluations.

3.5.7 Model Evaluation, Comparison, and Final Model Selection

Each model is trained and tuned, then compared against the held-out testing set by three standard regression metrics: the Coefficient of Determination (R^2), the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE). R^2 is the % of variance of the observed values that is accounted for by the predictions, RMSE and MAE are measures of prediction error, with lower values representing higher accuracy,

RMSE gives more weight to larger errors. Then, the models are compared and the model with the lowest RMSE, MAE and the highest R^2 is chosen as the final AQI forecasting model. When there are conflicting metrics, RMSE and MAE are given priority because the ability to make accurate predictions is crucial to the success of air quality forecasting for practical applications. The chosen model and the evaluation results are then continued in Chapter 4 for detailed analysis and the effectiveness of the Bayesian Optimization in boosting the model performance in terms of the optimized feature selection and hyperparameter tuning is indicated.

3.5.8 System Workflow Description

In Phase 2, the design is based on modules in a sequence creating an end-to-end machine learning pipeline, ranging from data input to preprocessing, Bayesian Optimization based feature selection, multi-model training, Bayesian Optimization based hyperparameter tuning, and evaluation. Four models are used: Random Forest, XGBoost, CatBoost and SVR, which allows for a comprehensive comparison, and a selection of traditional regression metrics (R^2 , RMSE, MAE) enables the best-performing forecasting model to be selected objectively. The design phase creates a stable baseline for the design and implementation of the architecture in Phase 3, which looks at the architecture and optimizes it with real world air quality data.

3.6 Phase 3: Implementation

Phase 3 consists of the implementation and operation of the AQI prediction system developed during Phase 2. The main purpose is to implement the conceptual architecture and process into a working machine learning system that generates accurate and reliable AQI forecasts. The data preprocessing, the building of several machine learning models and the determination of the best

possible features and hyperparameter combination with Bayesian optimization have been carried out in a well controlled and reproducible way. This phase acts as a bridge between design and evaluation by testing the proposed method with actual air quality data, and helps to get the system ready for performance testing and comparison in Phase 4.

3.6.1 Data Preprocessing Implementation

Experiments were conducted under a same environment so that the same outcomes can be reproduced for subsequent runs of the experiment. The preprocessing implementation consisted of the following tasks:

1) Handling missing values:

The rows missing a PM2.5 value were excluded without being imputed, because a supervised model needs a ground-truth target. The four air quality input features (NO₂, O₃, CO and PM10) were mean imputed because they had a small proportion of missing values which could be imputed without loss of valid readings.

2) Data type conversion and formatting:

Pollutant concentration variables were all converted to numeric format and date variables were standardized into a single datetime format for temporal consistency and sorting.

3) Duplicate and inconsistency handling:

Duplicate records and inconsistent entries, such as irregular spacing or invalid characters in numeric fields, were identified and corrected to prevent redundancy and bias during training.

4) Chronological ordering:

The data were stored in chronological order, which is more appropriate for data preprocessing, so that it is not sorted before it is stored. If shuffle the data before splitting into train and test sets,

then the model would be able to learn from the future observations as they are a part of the same time series.

5) Feature normalization:

Pollutant input was rescaled into a uniform range between 0 and 1 through min–max normalization, ensuring features are given a fair weight in learning process, especially in scale-sensitive algorithms like SVR.

Following the above steps, the dataset was now ready for feature selection, model training and hyperparameter optimization using Bayesian Optimization.

3.6.2 Feature Selection Implementation Using Bayesian Optimization

Following the pre-processing of acquired data, Bayesian Optimization was used to discover the set of air pollutant variables that are most relevant for AQI Prediction. Including irrelevant or redundant variables in the model may increase its computational complexity, slow down the generalisation and reduce its predictive accuracy, hence feature selection is crucial in the model development process (Elhariri et al., 2020).

The objective of this study was to use Bayesian Optimization to automate the selection process. Unlike brute force approaches (e.g. Grid Search) this probabilistic method allows to wisely balance exploration and exploitation, to help the algorithm to converge to high performing feature subsets while significantly reducing the computational cost (Harun et al., 2025).

The problem was completed in a wrapper-based approach in which the candidate sets of features were evaluated by training a machine learning model and assessing the performance based on some predefined objective

function. Metrics such as Root Mean Square Error (RMSE) and Coefficient of Determination (R^2) guided the optimization loop. The framework can be used to identify influential variables while preventing overfitting by adjusting the combinations of the features in an iterative process based on the historic evaluation results (Elhariri et al., 2020; Harun et al., 2025).

This optimized set of features is then used as the foundation for the subsequent stages of the training of the predictive system and hyperparameter tuning, aimed at training the predictive system with the most informative set of data available to it.

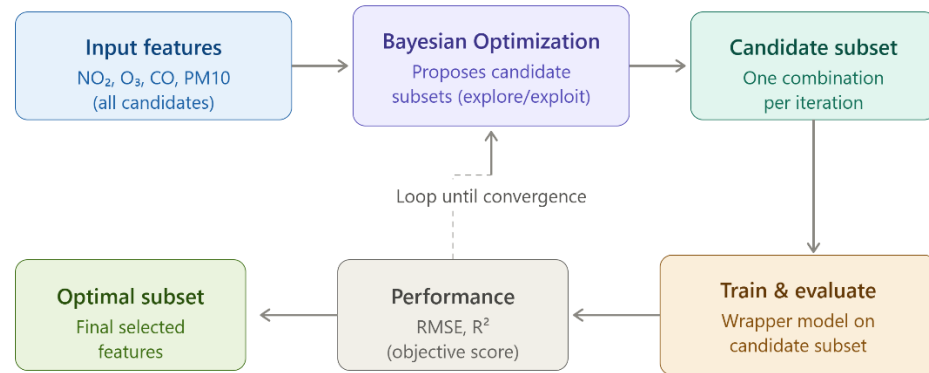


Figure 3.3 Workflow of feature selection using Bayesian Optimization

The workflow used for the feature selection through Bayesian Optimization is shown in Figure 3.3. The process begins at the complete level of air pollutant variables as the input. Bayesian Optimization then generates a subset of the features to find out the potential combinations of features by training a machine learning model and calculating a measure of prediction using RMSE and R^2 . The performance measures are then fed to the optimization engine to be used to improve the search approach step by step. This process is repeated until an optimal subset is found and taken to be used in the final training of the model and hyperparameter tuning.

3.6.3 Machine Learning Model Implementation

The models were trained on the selected features before hyperparameter tuning, and the four commonly used machine learning models were implemented to generate a baseline performance. The optimized feature subset extracted in the Bayesian Optimization stage of feature-selection was used as the same features for every model, which enabled the fair comparison of all algorithms. The dataset was split into a training and test set in a fixed chronological split for consistency and unbiased assessment, with the test set not used for training. For this purpose, four models were implemented:

1) Random Forest Regression:

Uses ensemble learning to reduce overfitting and model nonlinear relationships between air pollutant variables and AQI values.

2) XGBoost Regression:

Applies gradient boosting with regularization to achieve high predictive accuracy and robust generalization.

3) CatBoost Regression:

Captures complex feature interactions and reduces prediction bias through ordered boosting mechanisms.

4) Support Vector Regression (SVR):

Models nonlinear relationships using kernel functions and is particularly effective for high-dimensional regression problems.

All models were built with the standard Python machine learning libraries, and were trained using default hyperparameters, so that the learnability of each model could be assessed separately. The resulting forecasts on the test set are used as baseline results and compared following the standard regression metrics in Phase 4 and fed into the hyperparameter tuning phase.

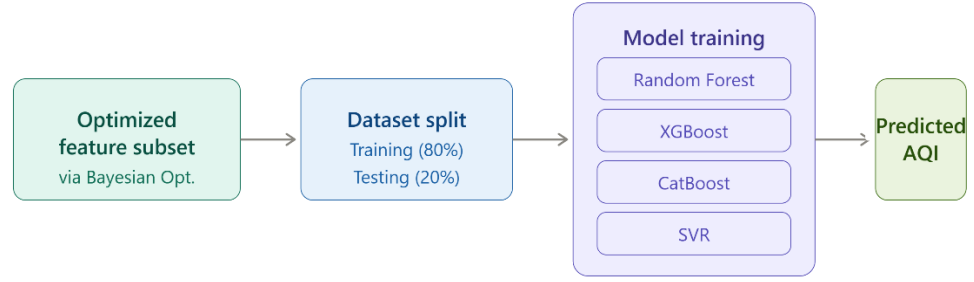


Figure 3.4 Machine learning model workflow

As illustrated in Figure 3.4, the optimized feature subset selected via Bayesian Optimization is split into training and testing sets. All four models (Random Forest, XGBoost, CatBoost and SVR) are trained on the training data and then predict the AQI values for the testing data which are evaluated in the testing stage.

3.6.4 Machine Learning Models

This section outlines a technical description of the machine learning algorithms used for the prediction of the Air Quality Index (AQI) for the current study. The evaluation criteria were evaluated considering the capacity of each model to describe the high disadvantage aspect from the dimension of the information entitled describing behavioural methods of high-dimensional data, as well as the capacity to represent the intricate interactions among the air pollutant concentrations, which were non-linear in nature (Babu & Kavitha, 2025).

3.6.4.1 Random Forest

Random Forest is an ensemble learning technique in which a large number of decision trees are trained and the regression prediction is taken to be the mean of the outputs of all the trees. It mitigates the overfitting effect and the variance of individual decision trees, by means of bagging (bootstrap aggregating) (Breiman, 2001). The advantages of using Random Forest in this study are its robustness to

noise and ability to capture the non-linear interactions among pollutant concentrations in air quality data. It also produces intrinsic feature importance metrics that are consistent with the project goal of selecting the most important pollutants. With the Bayesian-optimized feature subset, the model was instantiated to build a strong base-level predictor.

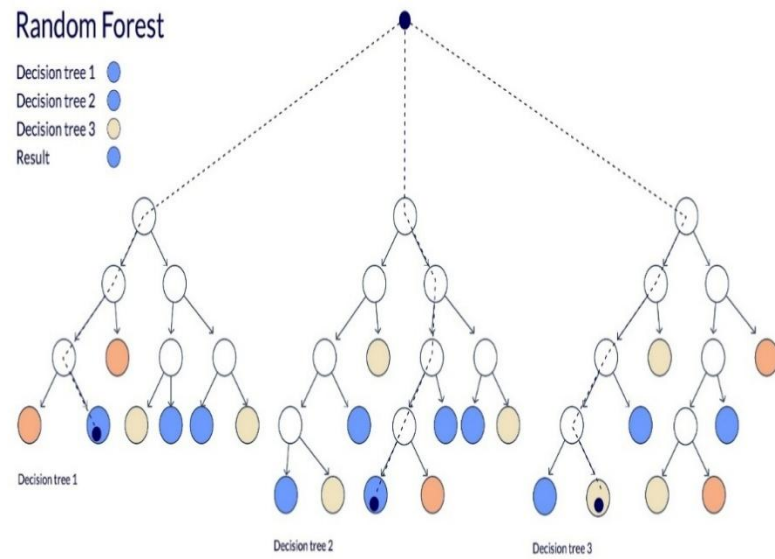


Figure 3.5 Random Forest for AQI prediction

(Source: Ravindiran et al., 2023)

The architecture of the Random Forest regression model is shown in figure 3.5. A series of bootstrap samples are first taken from the training set, and a separate decision tree is constructed for each sample, using a random subset of the predictor variables to determine splits. The final AQI prediction is computed using all the trees' predictions, for example, the mean of the arithmetic means. This ensemble approach generalizes and is more robust to prediction than single-estimator models.

In a regression task, the final prediction from the ensemble of decision trees (Random Forest) is a mean of all the individual decision tree predictions, as shown below (Breiman, 2001).

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (3.2)$$

where $\hat{y}$ is the predicted AQI value for input x , T is the total number of trees in the forest, and $f_t(x)$ is the prediction of the t -th tree. The averaging process helps minimize overfitting and generalizes well to unseen data, as it is well suited to the non-linear interactions among the concentrations of pollutants in the AQI data set (Breiman, 2001; Ravindiran et al., 2023).

3.6.4.2 XGBoost

XGBoost (Extreme Gradient Boosting) is a gradient boosting regression method which sequentially trains models, one model is correcting the error of the previous model and a built-in regularization makes the model simple and does not overfit. XGBoost is applied to model the non-linear association between the selected variables of air pollutants and AQI and is considered as a powerful benchmark because of its high accuracy, scalability and efficiency when applied to large historical datasets. The optimized feature subset was used to train the model which was then compared to other regression algorithms before hyperparameter optimization with Bayesian Optimization.

XGBoost uses an additive combination of K regression trees that, in effect, reduce the residual error of the previous trees, as illustrated below (Chen & Guestrin, 2016).

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (3.3)$$

where $\hat{y}_i$ is the predicted AQI value for the i -th instance, x_i is the vector of selected pollutant inputs, f_k is the k -th regression tree from the function space F of all possible trees, and K is the total number of trees.

One advantage of XGBoost is that each tree learns to minimise the residual prediction error, and a regularisation term is applied to the trees to prevent overfitting, thereby enabling XGBoost to model the nonlinear relationship between pollution and AQI with some confidence (Chen & Guestrin, 2016).

3.6.4.3 CatBoost

The CatBoost algorithm is a gradient boosting algorithm that works well with the complex relationships between features, requiring fewer feature interactions than other algorithms, and also allows for ordered boosting to reduce prediction bias, thereby reducing overfitting while retaining the algorithm's performance on different datasets. CatBoost is selected due to its ability to model complex relationships between the variables of air pollutants and AQI values and its robustness on nonlinear regression problems, making it suitable for environmental data. The optimized subset of features was used to train the model, which was then incorporated into the comparison of machine learning models.

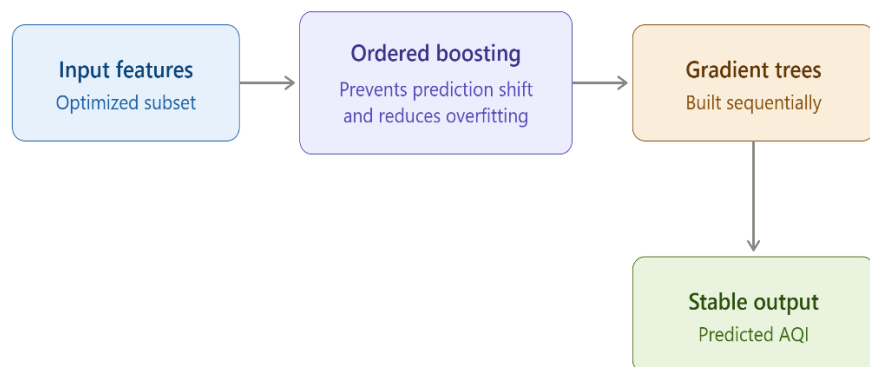


Figure 3.6 Conceptual illustration of CatBoost regression

(Source: Özüpak et al., 2025)

The workflow of the CatBoost regression model employed in this research is shown in Figure 3.6. The idea of ordered boosting is to prevent information leakage by producing the prediction for each training instance without relying upon the information gathered in other instances; this lowers bias, aside from the fact that it is a classical boosting method. This property makes CatBoost especially useful for structured, real-world environmental data, and enables it to maintain stable performance when handling data with varying distributions and relationships with pollutants in the AQI.

CatBoost train the model incrementally, adding a scaled base learner at every step, which is equivalent to the following (Prokhorenkova et al., 2018).

$$F_m(x) = F_{m-1}(x) + v \cdot h_m(x) \quad (3.4)$$

where $F_m(x)$ is the predicted AQI at iteration m , $F_{m-1}(x)$ is the prediction from the previous iteration, $h_m(x)$ is the base learner (tree) added at the m -th iteration, and v is the learning rate. By using ordered boosting, CatBoost reduces the prediction bias from the target leakage during training, which produces stable results in the different data distributions of the environment variables (Prokhorenkova et al., 2018; Özüpak et al., 2025).

3.6.4.4 Support Vector Regression

Support Vector Regression (SVR) is a kernel-based machine learning algorithm which can capture the nonlinear relationship in a high dimensional feature space and can minimize the prediction errors without over-fitting the model by regularizing it. SVR can process high dimensional data and can learn general associations between pollutants and AQI, which makes it suitable for AQI forecasting, and its kernel

functions can learn into nonlinear associations that are not so easy to learn using tree based methods. The optimum feature subset was then used for prediction of AQI by SVR and the results were compared to the ensemble-based models in this study.

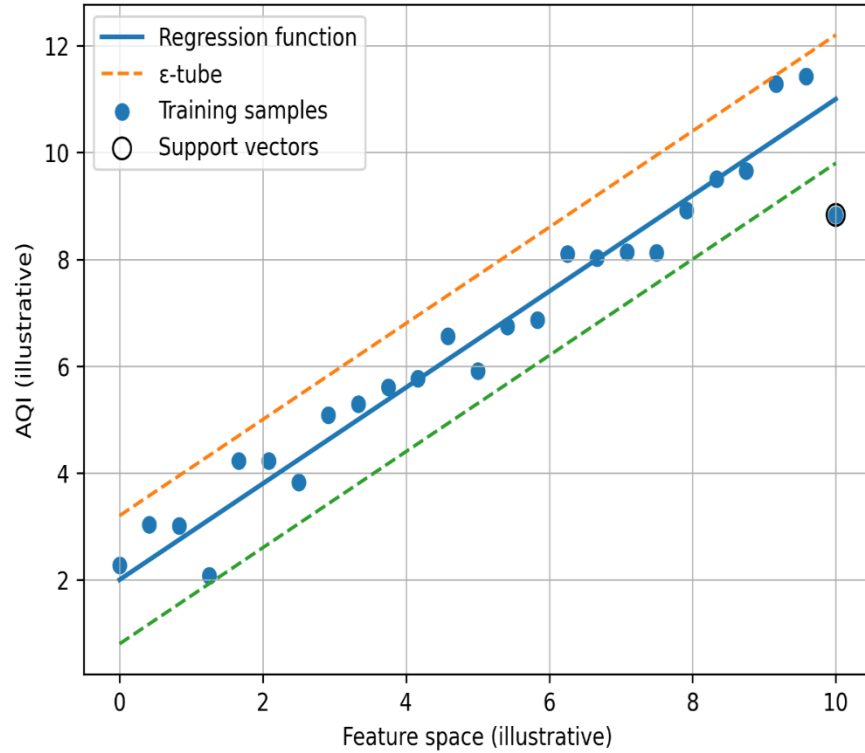


Figure 3.7 Conceptual illustration of Support Vector Regression (SVR)

(Source: Alwan et al., 2023)

The concept of SVR for prediction of AQI is shown in Figure 3.7. SVR is a model that fits a regression function to the data within some ϵ margin, but has a zero loss for errors less than ϵ from the true function. While the other ones (the so-called support vectors) do not affect the final regression function, SVR allows a balance between predictive capability and simplicity of the regression model. The optimized feature subset was used to train the SVR model, and the performance of prediction AQI was evaluated in comparison with ensemble models.

The training objective reduces the complexity of the model, as well as the ε -insensitive loss, which is defined below (Drucker et al., 1997).

$$\min_w \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \max(0, |y_i - f(x_i)| - \varepsilon) \quad (3.5)$$

where w is the weight vector, $(1/2)\|w\|^2$ is used to control the complexity of the model, C is the regularisation constant which balances the model complexity and training error, N represents the number of training samples, y_i represents the actual AQI value, $f(x_i)$ is the predicted AQI value, and ε is the width of the insensitive margin. The loss is ε -insensitive, which maintains the sparsity of the model, enabling SVR to model the nonlinear dependence of pollutants on AQI in high-dimensional feature space (Drucker et al., 1997; Alwan et al., 2023).

3.6.5 Hyperparameter Tuning Implementation Using Bayesian Optimization

Bayesian Optimization was then employed to optimize the hyperparameters of each model after implementation. Hyperparameter tuning is an important part of model development, as the performance of the developed model is sensitive to the quality of the selected hyperparameters and the purpose of the hyperparameter tuning phase is to find such configurations that will maximise the prediction accuracy and minimise the overfitting of the model.

Bayesian Optimization was chosen for its ability to efficiently explore complex, high-dimensional search spaces. Unlike grid or random search, it solves this problem by employing a probabilistic surrogate model to approximate the association of hyperparameter configurations to model

performance, thus exploring new regions of the search space while leveraging from existing high-performing hyperparameters.

The hyperparameters were tuned separately for each model on the chosen set of features. Each time the algorithm suggested a new configuration, trained a new model, and tested its performance and updated the surrogate model and adjusted the search. The hyperparameters that were tuned and their range of values were:

- 1) Random Forest (number of estimators [50–500], maximum tree depth [3–20], minimum samples per split [2–10])
- 2) XGBoost (learning rate [0.01–0.30], number of estimators [50–500], maximum tree depth [3–10], subsample ratio [0.6–1.0])
- 3) CatBoost (learning rate [0.01–0.30], number of boosting iterations [100–1000], tree depth [4–10])
- 4) SVR (kernel type, regularization parameter C, and epsilon).

The procedure was implemented using `gp_minimize` from the `scikit-optimize` (`skopt`) library, with a Gaussian Process surrogate model and the Expected Improvement (EI) acquisition function to balance exploration and exploitation. All evaluations conducted for a model were conducted 30 times on each candidate configuration, and the mean MAE was returned as the objective to be minimised using five-fold cross-validation with `TimeSeriesSplit`. The optimized configurations were then used to retrain each model prior to final evaluation.

3.6.5.1 Cross-Validation Strategy

The chronologically ordered training set was used to perform time-series cross validation to obtain a reliable estimate of generalization performance without inducing overfitting in the optimization of the tuning parameters. For each Bayesian Optimization evaluation, a candidate configuration of the hyperparameters was evaluated using five-fold cross validation with the TimeSeriesSplit cross validation strategy that maintains the temporal order of the data.

TimeSeriesSplit divides the training data into five distinct temporal folds such that for each validation fold it will always follow the training fold in time. Each proposed candidate configuration by `gp_minimize` had its model trained on the portion of the training set of each fold and validated on the next validation fold, and the average MAE over the five folds was returned as the objective to be minimized. This way, hyperparameter selection could not be influenced by information from subsequent observation folds.

After the 30 Bayesian Optimization evaluations, the configuration with the minimum mean MAE over the 5 folds of TimeSeriesSplit was chosen and this configuration was retrained on all 2,045 records in the training set and then evaluated on the held-out chronological test set (512 records). Table 3.2 provides a summary of the training, validation and test strategy.

Table 3.2 Training, Validation, and Test Strategy Summary

	Training	Testing
Split (%)	80	20
Role	Model fitting, feature selection, and hyperparameter tuning via Bayesian Optimization with cross-validation	Final held-out evaluation of the trained model on unseen data
Approx. Records	2,045	512

3.6.6 Stacking Ensemble Implementation

This study uses a Stacking Ensemble (stacked generalization) approach in order to enhance the predictive accuracy and generalization capabilities. Stacking is fundamentally different from traditional model selection which selects one optimal algorithm, as it trains multiple different algorithms (base learners) to extract different patterns from the data and then combines their predictions to enhance model performance and minimize the bias–variance trade-off (Zhang et al., 2021). The building has two levels.

For each of the four optimized models used in Section 3.6.5 (Random Forest, XGBoost, CatBoost and SVR), the model is trained independently on the training data with the hyperparameters found from the Bayesian Optimization. The model predictions are then obtained on the training set by means of 5-fold cross validation, which form the meta-features.

At Level-1 (meta-learner), the base-model predictions are aggregated to form a new meta-feature dataset. The meta-learner is a gradient-boosting model (`n_estimators = 200`, `learning_rate = 0.05`, `max_depth = 4`) of XGBoost. The original features input into the system are fed to the meta-learner with the base predictions to enable the meta-learner to adjust the base predictions with the raw pollutant inputs and to find the optimum way to combine the base predictions with the raw pollutant inputs to generate the final prediction for PM_{2.5}.

The overall procedure is as follows: The dataset is divided into training and testing data sets and then the four base learners are trained on the training data set; each base learner makes out-of-fold predictions on training data set (using 5-fold cross-validation), and along with the original features, the base learner predictions constitute the Level-1 data set; finally, the meta-

learner is trained on the Level-1 data set for the final prediction of AQI value. This hierarchical approach enables the end model to leverage the strengths of each individual algorithm, including the ability of XGBoost to learn in a non-linear way and the resilience of Random Forest, resulting in an enhanced forecasting model (Özüpak et al., 2025).

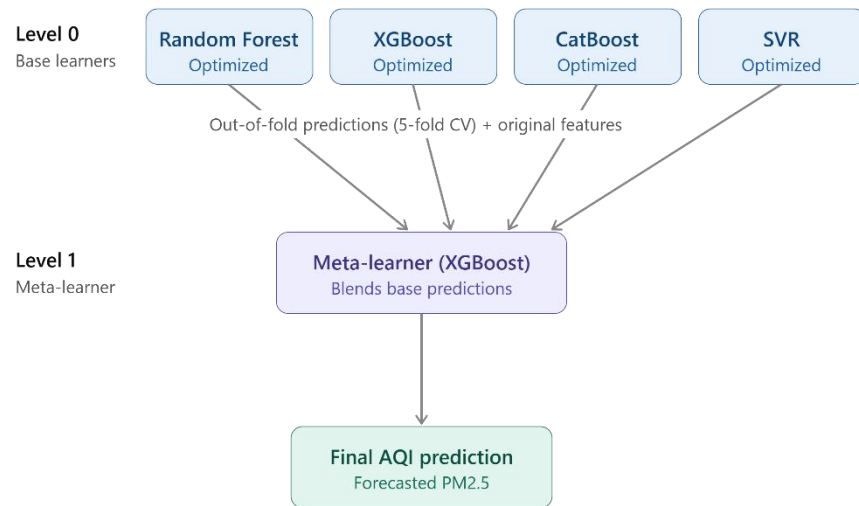


Figure 3.8 Two-level Stacking Ensemble architecture for AQI prediction

Figure 3.8 illustrates the two-level stacking ensemble architecture used in this study. At Level 0, the four optimized base learners (Random Forest, XGBoost, CatBoost and SVR) are trained independently on the training set and generate out-of-fold predictions using 5-fold cross-validation. These predictions, together with the original pollutant input features, form the meta-feature set passed to Level 1, where an XGBoost meta-learner blends them into the final forecasted PM2.5 value from which the AQI is derived. This hierarchical design allows the ensemble to draw on the complementary strengths of each base learner rather than relying on any single algorithm.

3.6.7 Implementation Summary

Phase 3 implemented the complete AQI forecasting system according to the architecture and workflow defined in this chapter, bringing the

methodology to life as a functional system in a systematic and reproducible manner. The implementation involved data preprocessing to clean and standardize the raw data, feature selection using Bayesian Optimization in order to select the most informative subset of pollutants to reduce dimensionality, the training of four machine learning models on the selected pollutant subset, and hyperparameter tuning using Bayesian Optimization, in order to optimize predictive performance while minimizing overfitting. Finally, a stacking ensemble of the individual models was used to combine the strengths of each model, and to increase the overall robustness. These allow for a robust and reliable foundation for Phase 4, where the predictive capabilities of the models are systematically assessed and compared, enabling the most suitable model to be used for AQI forecasting.

3.7 Evaluation Metrics

Quantitatively evaluate the predictive ability of the models created, three classical regression metrics were used: the Root Mean Squared Error (RMSE), the Mean Absolute Error (MAE) and the Coefficient of Determination (R²). The above metrics were chosen because they are the most commonly reported metrics in the air quality prediction literature, and thus the findings of this study can be readily compared with other published PM_{2.5} forecasting studies.

The MAE is the average absolute prediction errors irrespective of the direction. It is an average of the size of the errors and is less sensitive to outliers as each error is used proportionately to the others. It is reported in the same units as PM_{2.5} (µg/m³) and can therefore be directly interpreted as the average error between the predicted and observed concentration. The definition of MAE is given in Equation (3.1):

$$MAE = \frac{\sum_{j=1}^n |y_i - x_i|}{n} \quad (3.1)$$

The RMSE is used to assess the consistency of the model because it is more affected by large differences between the model and reality. The errors are squared before being averaged, which means that larger errors are given greater weight than smaller errors; therefore, RMSE is a good measure of the reliability of a model during periods of pollution spikes when larger errors are most likely to affect practical outcomes to protect public health. A smaller RMSE suggests that the model has less tendency to make big mistakes. RMSE is defined as below:

$$RMSE = \left[\sum_{j=1}^N (d_{fi} - d_d)^2 / N \right]^{\frac{1}{2}} \quad (3.2)$$

R^2 is the measure of the goodness of fit of the model and indicates the percentage of variation in the dependent variable (observed PM2.5 concentrations) accounted for by the independent variables. The value ranges from 0 to 1, with a value closer to 1 meaning that the model explains almost all the variation in the target variable, and a value closer to 0 meaning that the model does not do any better than predicting the mean of the target variable. R^2 is dimensionless, which means that the performance of the model can be compared without considering the scale of the pollutant concentration. R^2 is defined as in Equation (3.3):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.3)$$

where y_i denotes the observed PM2.5 value, $\hat{y}_i$ denotes the predicted value of PM2.5, $\bar{y}$ denotes the mean of the observations in the testing set, and n denotes the number of observations in the testing set.

All four base models (Random Forest, XGBoost, CatBoost, and Support Vector Regression) along with the stacking ensemble model were tested on the same chronologically held-out testing set using all three metrics. Using all three metrics together gives a well-rounded evaluation: MAE gives the typical magnitude of the error, RMSE reveals the model's performance when the error is large, and R^2 provides the overall goodness of fit. Therefore, a model is considered superior if it has lower RMSE and MAE values and higher R^2 value, and comparison between the individual models and the proposed stacking ensemble is theoretically and practically meaningful for the practical use of air quality monitoring in the real world.

3.8 Phase 4: Testing

Phase 4 tests the predictive ability of the developed AQI prediction system and ensures it fulfils the project requirements and provides forecasts that will be accurate, reliable and generalizable. During this stage, the trained models are evaluated using unseen testing data to validate their generalization performance under realistic air quality conditions, which is free of human interference. During this phase, the trained models are tested with unseen testing data to validate their generalization performance in the real air quality status free of human interference.

The evaluation of the models is done using three common regression indicators: Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Coefficient of Determination (R^2). R^2 value represents the percentage of variance in observed AQI values that is explained by the predicted values and RMSE and MAE are used to measure the size of the prediction error, with lower values being more accurate. All the performance is subsequently compared to each other to determine the most effective AQI prediction model and the effects of the Bayesian Optimization are evaluated by comparing the performance before and after optimized feature selection and

hyperparameter tuning. The results of this evaluation are detailed and discussed in Chapter 4.

3.8.1 Model Performance Evaluation and Comparison

The evaluation and comparison of the machine learning models applied to AQI prediction is described in this subsection. The goal is to use quantitative measures to evaluate each model's predictive performance and generalization capabilities to assist in selecting the most appropriate model (Babu & Kavitha, 2025).

The R^2 , RMSE, and MAE values are calculated for each trained and tuned model to evaluate the model on held out test data. The R^2 values are higher when the explanatory power is higher and lower when the prediction of the AQI is more accurate and stable. The higher the R^2 values the better the explanatory power and the lower the RMSE and MAE values the more accurate and stable the prediction of AQI. The best performing individual model is identified as the one with the highest R^2 , lowest RMSE and lowest MAE. When metrics compete, RMSE and MAE have priority as minimising prediction error is fundamental for air quality forecasting use. This comparative method allows for unbiased selection of models and lays the foundation for the findings in Chapter 4.

3.8.2 Impact of Bayesian Optimization on Model Performance

This subsection provides details of the assessment of the effect of Bayesian Optimization on model performance. The proposed system includes a two-stage Bayesian optimization process: first, for feature selection, to enhance the predictive performance and resistance to overfitting; second, for hyperparameter tuning, to boost the model's performance and adaptability.

The impact is measured using the same regression metrics (MAE, RMSE and R^2) used before and after optimization of the models. Higher R^2 and lower RMSE and MAE are signs of performance improvement due to better explanatory power and smaller RMSE and MAE. The feature selection technique is applied to reduce the number of input variables by selecting the most relevant pollutant variables among the five, minimizing noise and maximizing learning efficiency, while the hyperparameter tuning technique optimizes the settings of each model to minimize risk of overfitting it. In Chapter 4, the quantitative results of this comparison are presented.

3.8 Summary

The research methodology adopted in the development of the proposed AQI forecasting system was explained in this chapter. The process was structured and systematic in the Waterfall SDLC model with four (4) phases including Requirement Analysis, System Design, Implementation and Testing, thus enabling the research to be carried out in a systematic and structured manner.

Phase 1 identified the system requirements and the process of data collection was determined based on air quality data from Kuching, Sarawak region. Phase 2 included the overall system design that set out the modular architecture and workflow for the AQI prediction. In Phase 3, the implementation process involved data preprocessing, feature selection and hyperparameter optimization using Bayesian Optimization, training of four machine learning models (Random Forest, XGBoost, CatBoost and SVR), and ensembling these models to achieve better results. Phase 4 described the test procedure in which the models were tested on unseen data and evaluated based on well-known regression metrics R^2 , RMSE and MAE to determine the accuracy and the generalization ability of the models.

Overall, this structured method is robust and repeatable to build the AQI forecasting system according to the research goals. The findings and results of this implementation are presented and discussed in chapter 4.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results and findings of the project titled Improving Air Quality Prediction Using Bayesian Optimization for Feature Selection and Hyperparameter Tuning. It provides an overview of the project system, including the dataset characteristics, data pre-processing, feature selection, and the baseline and optimized performance of the machine learning models. The performance of the stacking ensemble and the accuracy of the resulting Air Quality Index (AQI) classification are then evaluated, followed by a discussion of the prototype development.

4.1 Introduction

The experimental results of the suggested air quality prediction framework are discussed for the city of Kuching, Sarawak, Malaysia in this chapter. The framework combines the Bayesian optimization feature selection and hyperparameter tuning techniques with a stacking technique for ensemble learning. The results showed the dataset properties, pre-processing results, feature selection, baseline and optimized model performance, stacking ensemble results and accuracy of the Air Quality Index (AQI) forecast.

The key performance indicators include the coefficient of determination (R^2), the Mean Absolute Error (MAE), and the Root Mean Squared Error (RMSE). When a model has a $R^2 \geq 0.80$, which means that 80% or more of the variance in PM2.5 concentration is explained by the predictor variables, it is considered a satisfactory model. The category match accuracy is an additional application-level metric used to show the usefulness of predictions for public health advisories. In this updated model, PM2.5 is predicted in a multivariate model with four input pollutants (NO_2 , O_3 , CO , PM_{10}) and without the use of lag-based features to be achieve this objective.

4.2 Dataset Overview

Five air quality parameters (NO₂, O₃, CO, PM10 and PM2.5) are measured daily in Kuching. The multivariate input features were the remaining four pollutants, PM2.5 was the prediction target. The characteristics of the data set after pre-processing are presented in Table 4.1.

Table 4.1 Dataset Characteristics for Kuching

Characteristic	Kuching
Total observations	3,026
Missing PM2.5 readings (dropped)	469 ($\approx 15.5\%$)

The Kuching dataset comprised of 3,026 daily records but had 469 missing PM2.5 readings (about 15.5% of the data) in addition to a few zero readings for the input pollutants, Figure 4.1 Overview of Dataset is overview data from 1-20.

1	Site_Id	Site_Location	Date	no2_hourly_avg	O3_hourly_avg	Co_hourly_avg	pm10_hourly_avg	1
2	CA0004	Medical Store, Kuching	20160101	0.004	0.011	0.447	23.667	
3	CA0004	Medical Store, Kuching	20160102	0.003	0.013	0.440	21.625	
4	CA0004	Medical Store, Kuching	20160103	0.003	0.013	0.399	12.083	
5	CA0004	Medical Store, Kuching	20160104	0.005	0.011	0.503	39.167	
6	CA0004	Medical Store, Kuching	20160105	0.006	0.009	0.571	32.542	
7	CA0004	Medical Store, Kuching	20160106	0.006	0.009	0.541	26.667	
8	CA0004	Medical Store, Kuching	20160107	0.005	0.011	0.500	28.042	
9	CA0004	Medical Store, Kuching	20160108	0.005	0.012	0.428	20.875	
10	CA0004	Medical Store, Kuching	20160109	0.007	0.010	0.553	26.167	

Figure 4.1 Overview of Kuching Dataset

4.3 Data Preprocessing

The raw data of air quality data for Kuching has been introduced in Section 4.2 and the data preprocessing procedures outlined in Chapter 3 (Section 3.4) were thus implemented upon the raw data. The result of each preprocessing stage in

data cleaning, missing values, feature-target definition, chronological train-test splitting and normalization is presented and the characteristic of the resulting dataset is presented and carried forward to feature selection and model training in Sections 4.4 to 4.6.

4.3.1 Data Cleaning

Before further processing, the raw dataset was first cleaned to correct error in structure and encoding errors. The `drop_duplicates()` function was used to remove any duplicate rows if present, which could be a result of the sensor logging multiple times. All pollutant columns (NO_2 , O_3 , CO, PM10 and PM2.5) were then converted to a numeric data type using `pd.to_numeric()` with the `errors='coerce'` argument, meaning that the pipeline would not fail if some pollutant data was not numeric or corrupted.

Some pollutants like PM2.5 and PM10 are not able to achieve a true zero reading in ambient air quality sensors, and a true zero reading is more likely to be a sensor fault. To address this, all the readings in the five pollutant columns were filled in with NaN to give the same treatment to erroneous readings as to other missing data in the subsequent steps. Figure 4.2 shows the implementation of these cleaning operations.

```
df = df.drop_duplicates().reset_index(drop=True)
pollutant_cols = ['NO2', 'O3', 'CO', 'PM10', 'PM2.5']
for col in pollutant_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
    df[col] = df[col].replace(0, np.nan)
```

Figure 4.2 Code snippet of duplicate removal, numeric type coercion and zero-value handling

4.3.2 Handling of Missing Values

After cleaning as described in Section 4.3.1, there were two types of missing values in the data: missing values in the target variable, PM2.5, and missing values in the four input features used for pollutants (NO₂, O₃, CO, PM10). As per the strategy outlined in Chapter 3, both categories were handled separately.

If the target variable was missing from a row of the dataset, then it could not be used in supervised learning, because there would be no ground truth PM2.5 concentration to which a prediction could be compared. The rows were therefore dropped from the data set rather than imputed as demonstrated in Figure 4.3 Applying this rule resulted in the removal of 469 rows from the original 3,026-row data set (approximately 15.5%). This was in line with the amount of missingness identified during the raw data quality review. This indicates that the 469 readings that were reported as being affected were dropped, not imputed, as no replacement value was generated for the missing target readings.

```
df = df.dropna(subset=['PM2.5']).reset_index(drop=True)
```

Figure 4.3 Code snippet removal of rows with missing PM2.5 target values

The other four features with missing values were filled with the column mean, however. Compared with the target variable, these features are not affected by small percentage of missing values and mean imputation does not discard valid PM2.5 readings if there are missing values on the same row. Following this, there were no missing values in the data anywhere.

4.3.3 Feature and Target Definition

Given the complete and cleaned dataset, the dataset consisting of the four remaining pollutant variables (NO₂, O₃, CO and PM10) was defined as the input feature matrix X and the target vector y was defined as PM2.5, as

shown in Figure 4.4. This four-feature configuration is in line with the feature set found in Chapter 3 and is subsequently verified through the use of a Bayesian Optimization feature selection procedure as detailed in Section 4.4.

```
feature_cols = ['NO2', 'O3', 'CO', 'PM10']  
X = df[feature_cols].copy()  
y = df['PM2.5'].copy()
```

Figure 4.4 Code snippet definition of the input feature matrix (X) and target vector (y)

4.3.4 Chronological Train-Test Splitting

For the partitioning, chronologically sorted the cleaned data set and then used an 80/20 split as shown in Figure 4.5, where the first 80% of data in date order was considered the training set and the last 20% was considered the testing set. There were 2557 cleaned records; 2045 of which were used for training and 512 for testing.

```
split_idx = int(len(X) * 0.8)  
X_train_raw = X.iloc[:split_idx].reset_index(drop=True)  
X_test_raw = X.iloc[split_idx:].reset_index(drop=True)  
y_train = y.iloc[:split_idx].reset_index(drop=True)  
y_test = y.iloc[split_idx:].reset_index(drop=True)
```

Figure 4.5 Chronological 80/20 train-test split

A split that happened chronologically was used instead of a random split, as the data set is a time-ordered series of daily observations of air quality. Random shuffling would make it possible to train the model on records that were collected beyond the period of the prediction and therefore would not be reflective of how the model would be applied in an actual forecasting situation, that is, on past observations to predict future air quality. Data splitting was done prior to normalization (which is required to avoid data leakage as described in Section 4.3.5).

4.3.5 Data Normalization

Min–Max normalization was applied to rescale PM10 to the range [0, 1], as shown in Figure 4.6.

```
scaler = MinMaxScaler()  
X_train = X_train_raw.copy()  
X_test = X_test_raw.copy()  
X_train[['PM10']] = scaler.fit_transform(X_train_raw[['PM10']])  
X_test[['PM10']] = scaler.transform(X_test_raw[['PM10']])
```

Figure 4.6 Code snippet Min-Max normalization

Only the feature "PM10" was normalized because it is the only feature that varies by orders of magnitude from the others. Within the training partition, PM10 spans 5.734 to 237.367 $\mu\text{g}/\text{m}^3$, while NO₂ (0.0008–0.0140), O₃ (0.0006–0.0344) and CO (0.082–3.308) are already confined to narrow, comparable ranges. Rescaling PM10 is thus important for SVM, as kernel methods are based on the relative magnitude of features and PM10 would otherwise have an overwhelming influence on the kernel, even if it were not a strong predictor. The tree-based learners are scale-invariant and are unharmed by this.

The scaler was applied only to the training set to prevent data leakage, and the minimum and maximum values of the training set were used to transform the testing set. It is not realistic in a real forecasting situation for the distribution of future observations to affect the scaling of the combined data, as would be done in the fitting of the combined or testing data. Table 4.2 shows a sample of PM10 data before and after normalization.

Table 4.2 Sample of Feature Before and After Normalization

Feature	Original value	Normalized value
PM10	23.667	0.0774

4.3.6 Exploratory Data Analysis (EDA)

Before feature selection and model training, the cleaned data underwent exploratory data analysis to analyze the distributions of the variables, detect outliers, and assess the relationships between pollutants.

The missingness of the raw and the cleaned data was compared first (see Figure 4.3). The four input pollutants had very little missing data (< 1%) and PM2.5 had 15.50% missing, which is the 469 rows that were removed in Section 4.3.2. After cleaning no missing values were left. A descriptive analysis of the resulting 2,557-record data set is given in Table 4.3.

Table 4.3 Descriptive statistics of the cleaned dataset

Variable	Mean	Std. Dev.	Skewness
NO ₂	0.0052	0.0019	0.45
O ₃	0.0120	0.0041	1.14
CO	0.4880	0.2571	2.61
PM10	19.73	15.42	7.15
PM2.5	12.22	13.92	7.78

The histograms in Figure 4.7 show that NO₂ is approximately symmetric (skewness 0.45), while O₃ is moderately right-skewed (1.14). CO, PM10 and PM2.5 are strongly right-skewed (2.61, 7.15 and 7.78 respectively), with most readings concentrated at low concentrations and a long tail of occasional high-pollution episodes. This pronounced skew in the particulate matter variables reflects the episodic nature of haze events in the region and justifies the use of tree-based and ensemble learners, which handle non-normal feature distributions without requiring transformation.

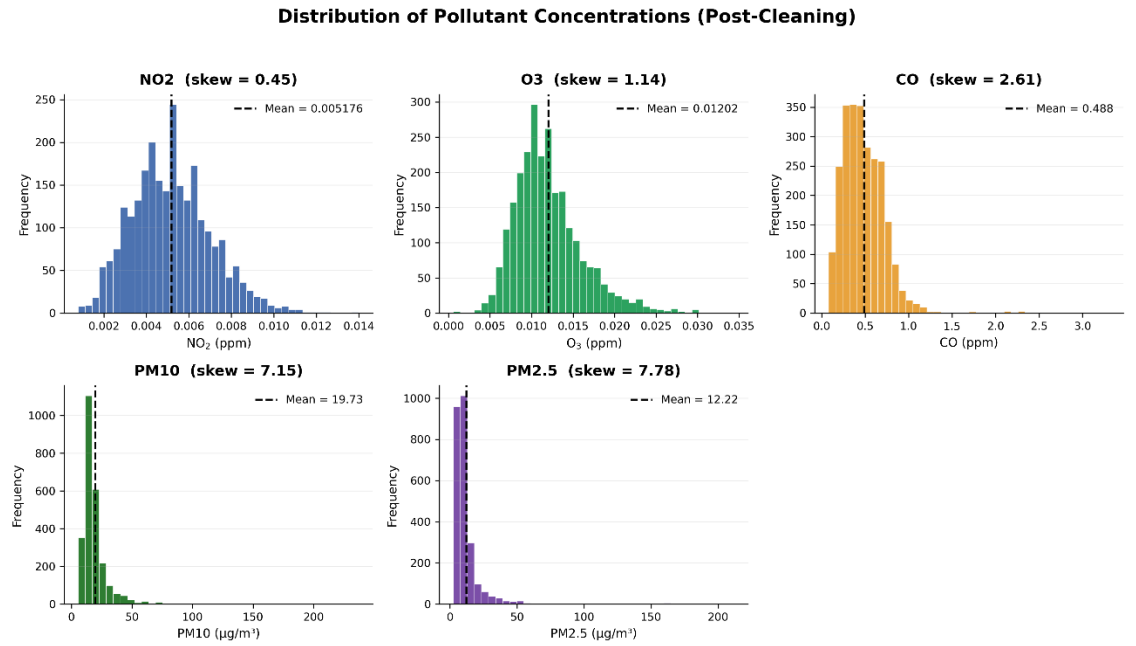


Figure 4.7 Distribution of Air Pollutant Variables After Data Cleaning

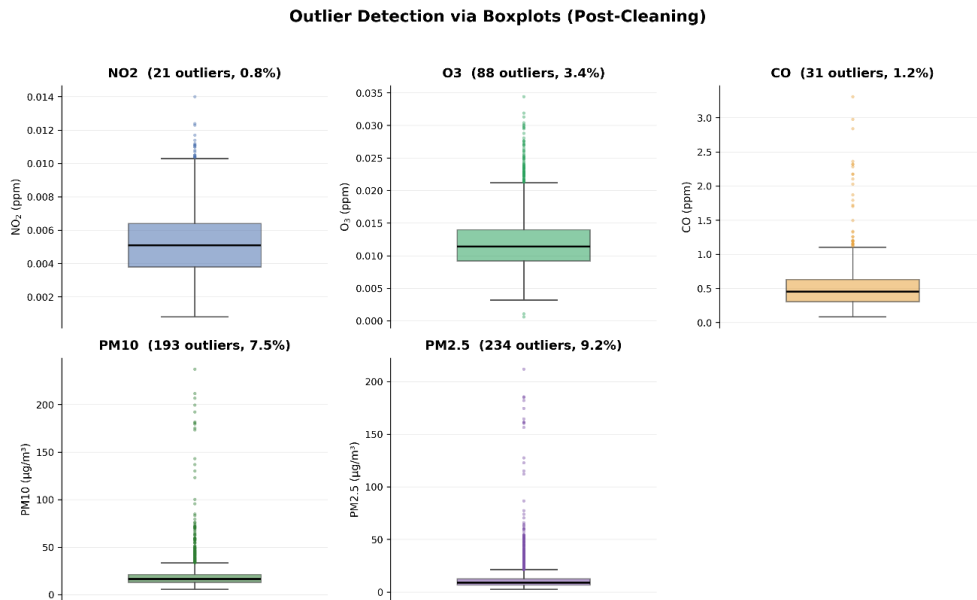


Figure 4.8 Outlier Detection After Data Cleaning

Boxplots of the data for the five pollutant variables after cleaning are shown in Figure 4.8. The Outliers were determined by the criterion of the Interquartile range (IQR), which is a range from the values of $Q1 - 1.5 \times$

IQR to $Q3 + 1.5 * IQR$. There are relatively few extremes for the gaseous pollutants, 21 (0.82%) for NO_2 , 31 (1.21%) for CO and 88 (3.44%) for O_3 . The high positive skewness reported in Table 4.3 is confirmed by the particulate variables as they present much greater number of outliers with 193 (7.55%) for PM10 and 234 (9.15%) for PM2.5.

While these are extreme values, they have not been excluded as they are a true representation of actual high pollution events e.g. haze events, and not measurement errors. By discarding them, they would remove very exactly the conditions the model requires to predict, and the ensemble learners used in this study are robust to such values.

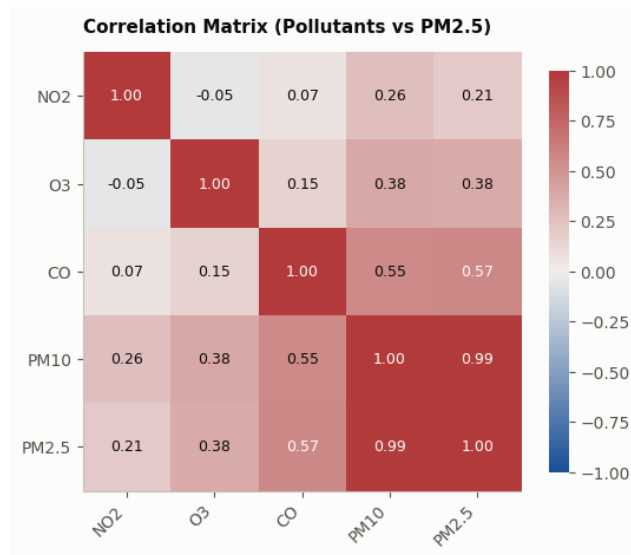


Figure 4.9 Correlation heatmap

A correlation heatmap as shown in Figure 4.10 revealed that there is very high correlation between PM10 and PM2.5 ($r = 0.986$) as would be expected because PM2.5 is part of PM10. CO showed a moderate correlation ($r = 0.566$), while O_3 ($r = 0.379$) and NO_2 ($r = 0.208$) were weaker. PM10 was still used as a feature at this point, as actual feature selection will be performed later in Section 4.4, based on Bayesian Optimization and not on correlation.

Boxplots and interquartile range (IQR) method were also used to explore outlier prevalence, which sets an observation as an outlier if it is more than 1.5 times the IQR from either quartile, as illustrated in Figure 4.8. The resulting outlier proportions were 0.82% for NO₂, 3.44% for O₃, 1.21% for CO, 7.55% for PM10 and 9.15% for PM2.5. These observations were not discarded but rather kept for analysis, as they are real data from high pollution events (not measurement error) and their exclusion could impact the model's ability to predict hazardous air quality events that are most relevant to this study; consistent with the heavy right-skew of PM10 and PM2.5, these events were retained rather than removed from the data set.

4.4 Feature Selection via Bayesian Optimization

Feature selection was done by a direct search over a subset of the input features, based on a Gaussian Process surrogate model and the Expected Improvement (EI) acquisition function. The goal was to get the minimum cross validated mean squared error across 5 folds, using a Random Forest regressor. A search was conducted for 30 evaluations. A continuous selection score (from 0 to 1) was assigned for each feature, and a feature was selected to a subset when its continuous score was higher than 0.5. The selection scores obtained for the Kuching data set are shown in Table 4.4.

Table 4.4 Feature Selection Scores for Kuching dataset

Feature	Selection Score	Selected (Score > 0.5)
PM10	0.6833	Yes
CO	0.2410	No
O ₃	0.2309	No
NO ₂	0.0160	No

CO (0.2410), O₃ (0.2309) and NO₂ (0.0160) were all below the inclusion threshold while PM10 obtained the highest selection score (0.6833) and is the only feature that is above the threshold. The high concentration of PM10 is consistent with the high physical linkage between the particulate matter fractions: PM10 and PM2.5 have similar emission sources, such as vehicular traffic, biomass burning and industrial activity.

A fall-back condition was added to ensure the ability to include fewer than two features while maintaining the interpretability and robustness of the model. This condition was encountered because PM10 is the only feature to pass, so the entire feature set was kept. For this reason, all the four pollutants (NO₂, O₃, CO and PM10) were employed for the training of the models.

4.5 Baseline and Optimized Model Performance

This section evaluates the four base learners Random Forest, XGBoost, CatBoost, and Support Vector Regressor in two configurations. Each is first trained in a baseline form using default hyperparameters, then re-trained in an optimized form with its key hyperparameters tuned through Bayesian optimization, isolating the effect of tuning from the choice of algorithm.

Each baseline model and learning mechanism is described in Sections 4.5.1 to 4.5.4, and the optimized model and the improvement achieved by each model is presented in Section 4.5.5. All models are tested with the chronological 80/20 split (2,045 training records, 512 test records) and the same test set is used for evaluation of the models by RMSE, MAE and R².

4.5.1 Random Forest

Random Forest is a bagging ensemble (average of many decision trees) which is robust against overfitting and noise. There are only two hyperparameters set in the baseline instantiation, namely the parameter

random_state=42 (to ensure the results are reproducible) and the parameter n_jobs=-1 (for the parallel training) by based on Figure 4.11 code snippet below:

```
('Random Forest (Default)',  
RandomForestRegressor(random_state=42, n_jobs=-1)),
```

Figure 4.11 Code snippet Random Forest

4.5.2 XGBoost

The XGBoost is an implementation of a gradient boosting framework in which trees are added in sequence with l2 regularization to prevent overfitting. Figure 4.12 demonstrates the baseline XGBRegressor: random_state=42 defines the seed, objective='reg:squarederror' specifies the regression task, and verbosity=0 turns off the training log, while the learning rate, depth and number of rounds remain at default.

```
('XGBoost (Default)',  
XGBRegressor(random_state=42, objective='reg:squarederror', verbosity=0)),
```

Figure 4.12 Code snippet XGBoost

4.5.3 CatBoost

CatBoost is a fast and efficient gradient boosting implementation to process feature interactions and with minimal tuning. Only the random_state=42 (reproducibility) and verbose=0 (silences per-iteration output) are changed in the baseline CatBoostRegressor shown in Figure 4.13 the learning rate, depth and iterations are kept at their default values.

```
('CatBoost (Default)',  
CatBoostRegressor(verbose=0, random_state=42)),
```

Figure 4.13 Code snippet CatBoost

4.5.4 Support Vector Regressor (SVR)

As a contrast to the tree-based ensembles, Support Vector Regression (SVR) is a kernel-based technique that fits a function within a tolerance margin. The baseline is a call to the SVR function with no arguments (SVR doesn't have a `random_state`), default RBF kernel and default `C`, `epsilon` and `gamma` as shown in Figure 4.14 Being highly sensitive to these settings, this untuned baseline is expected to be the weakest of the four.

```
('SVR (Default)',  
SVR()),
```

Figure 4.14 Code snippet SVR

4.5.5 Optimized Model Performance

This phase optimises the hyperparameters of each of the 4 models using Bayesian Optimization and compares the optimised models to the baseline, measuring how much better each is.

4.5.5.1 Hyperparameter via Bayesian Optimization

In this section, used of Bayesian optimization to find a better configuration for each algorithm. The same tuning procedure (indicated in each subsection) is repeated using `gp_minimize` from `scikit-optimize`, using the Expected Improvement acquisition function, which uses five-fold cross-validation based on the

TimeSeriesSplit to evaluate each candidate, and returns the mean MAE as the objective. Each search was conducted over 30 evaluations, and a final model was refitted using the best parameters found in the search on the complete training set. Each algorithm's search space and best configuration is provided in the subsections below.

4.5.5.1.1 Random Forest (Bayesian Optimize)

In the case of Random Forest, `n_estimators` (100-500), `max_depth` (3-20), `min_samples_split` (2-10), and `max_features` (0.3-1.0) were optimized. The search is shown in Figure 4.15, Hyper-parameter optimisation to the best values yielded `n_estimators` = 100, `max_depth` = 7, `min_samples_split` = 2 and `max_features` = 1.0, resulting in an increase in the test R^2 from 0.9224 to 0.9339.

```
print('Tuning Random Forest...')
rf_params = bo_tune(
    RandomForestRegressor,
    [Integer(100, 500, name='n_estimators'),
     Integer(3, 20, name='max_depth'),
     Integer(2, 10, name='min_samples_split'),
     Real(0.3, 1.0, name='max_features')],
    {'random_state': 42, 'n_jobs': -1}
)
print(f' Best params: {rf_params}')
best_rf = RandomForestRegressor(**rf_params, random_state=42, n_jobs=-1)
best_rf.fit(X_train_sel, y_train)
```

Figure 4.15 Code snippet Random Forest hyperparameters with Bayesian optimization

4.5.5.1.2 XGBoost (Bayesian Optimize)

For XGBoost, the hyperparameters that were explored were `n_estimators` (50–500), `max_depth` (3–10), `learning_rate` (0.01–0.3), `subsample` and `colsample_bytree` (each 0.5–1.0), and `reg_alpha` (0.0–1.0). The effect of the learning rate was multiplicative and so

was represented by a log scale. The search is shown in Figure 4.16. The optimal was `n_estimators = 500`, `max_depth = 3`, `learning_rate` ≈ 0.113 , `subsample = 1.0`, `colsample_bytree = 1.0`, and `reg_alpha = 0.0` (many shallow trees), resulting in a test R^2 of 0.9010 compared to 0.8809 at baseline.

```
print('Tuning Random Forest...')
rf_params = bo_tune(
    RandomForestRegressor,
    [Integer(100, 500, name='n_estimators'),
      Integer(3, 20, name='max_depth'),
      Integer(2, 10, name='min_samples_split'),
      Real(0.3, 1.0, name='max_features')],
    {'random_state': 42, 'n_jobs': -1}
)
print(f' Best params: {rf_params}')
best_rf = RandomForestRegressor(**rf_params, random_state=42, n_jobs=-1)
best_rf.fit(X_train_sel, y_train)
```

Figure 4.16 Code snippet XGBoost hyperparameters with Bayesian optimization

4.5.5.1.3 CatBoost (Bayesian Optimize)

For CatBoost, four hyperparameters were tuned: `learning_rate` (0.01–0.3, log scale), `depth` (4–10), `iterations` (200–1000), and `l2_leaf_reg` (1.0–10.0). The search is shown in Figure 4.17, the most successful setting was `learning_rate` ≈ 0.058 , `depth = 4`, `iterations = 242`, `l2_leaf_reg = 1.0` with the test R^2 increasing from 0.9009 to 0.9332, the biggest improvement of the tree-based learners and the second-highest R square of all other models.

```

print('Tuning Random Forest...')
rf_params = bo_tune(
    RandomForestRegressor,
    [Integer(100, 500, name='n_estimators'),
      Integer(3, 20, name='max_depth'),
      Integer(2, 10, name='min_samples_split'),
      Real(0.3, 1.0, name='max_features')],
    {'random_state': 42, 'n_jobs': -1})
print(f' Best params: {rf_params}')
best_rf = RandomForestRegressor(**rf_params, random_state=42, n_jobs=-1)
best_rf.fit(X_train_sel, y_train)

```

Figure 4.17 Code snippet CatBoost hyperparameters with Bayesian optimization

4.5.5.1.4 Support Vector Regressor (Bayesian Optimize)

The three hyperparameters C (0.1 – 100), epsilon (0.001 – 1.0) and gamma (0.0001 – 1.0) were optimized, all on a log scale, because the SVR is sensitive to these parameters; fixed_kw is an empty object since there is no random_state to be optimized in the case of SVR. The search is shown in Figure 4.18. The best configuration was $C = 100$, epsilon = 0.001, and gamma ≈ 0.239 , producing the largest improvement of all four models: the test R^2 rose from 0.8405 to 0.9235, the RMSE fell from 3.6425 to 2.5217, and the MAE from 2.0316 to 1.6237.

```

print('Tuning Random Forest...')
rf_params = bo_tune(
    RandomForestRegressor,
    [Integer(100, 500, name='n_estimators'),
      Integer(3, 20, name='max_depth'),
      Integer(2, 10, name='min_samples_split'),
      Real(0.3, 1.0, name='max_features')],
    {'random_state': 42, 'n_jobs': -1})
print(f' Best params: {rf_params}')
best_rf = RandomForestRegressor(**rf_params, random_state=42, n_jobs=-1)
best_rf.fit(X_train_sel, y_train)

```

Figure 4.18 Code snippet SVR hyperparameters with Bayesian optimization

4.6 Comparison of Baseline and Optimized Models

Each algorithm was evaluated separately, both with default and Bayesian optimized configurations (Sections 4.5.1-4.5.5). This section combines those results to measure the overall impact of Bayesian Optimization on the four learners and to determine the best algorithm for each performance measure. The comparison is done for Objective 1 and 2 and serves as the empirical foundation for the stacking ensemble in Section 4.7. The results are shown separately for the training partition (Section 4.6.1) and the held-out test partition (Section 4.6.2).

4.6.1 Train-set

This section reports training-set performance for the four base learners before and after Bayesian Optimization across three metrics: MAE (Table 4.5), RMSE (Table 4.6), and R^2 (Table 4.7). The baseline column records each model's default hyperparameter configuration, the optimized column the configuration selected by Bayesian Optimization, and the improvement percentage the relative change between them.

Training-set results are diagnostic rather than evaluative. Since the optimization minimized cross-validated error rather than training error, a decline in training performance after tuning reflects intended regularization, not a failed search. These results should be read alongside the test-set results in Section 4.6.2, which provide the definitive assessment of generalization.

Table 4.5 MAE Improvement from Bayesian Optimization for Kuching
(Train Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	0.5345	1.2629	-136.28
XGBoost	0.4273	0.9317	-118.04
CatBoost	0.8997	1.2629	-40.37
SVR	0.8997	1.5652	45.54

Table 4.5 RMSE Improvement from Bayesian Optimization for Kuching
(Train Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	0.8277	1.6959	-104.89
XGBoost	0.5892	1.2302	-108.79
CatBoost	1.1685	1.7108	-46.41
SVR	11.3489	2.2872	79.85

Table 4.6 R² Improvement from Bayesian Optimization for Kuching
(Train Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	0.9969	0.9870	-0.99

Model	Baseline MAE	Optimized MAE	Improvement (%)
XGBoost	0.9984	0.9932	-0.52
CatBoost	0.9938	0.9868	-0.70
SVR	0.4179	0.9764	133.64

On training set, the three tree based learners are worse after tuning on this set, and SVR improves drastically on this set. This is a desirable and expected behaviour, not an optimization failure. Random Forest, XGBoost and CatBoost with default parameters fit almost perfectly (R^2 of 0.9969, 0.9984 and 0.9938 respectively) the training data, typical of unconstrained tree ensembles memorising the training set. For example, when using Random Forest, they set the max depth to 6, and XGBoost to 50 estimators at depth 3, compromising the fit of the training set in exchange for better generalization. The opposite occurred for SVR, as it was severely underfitting the data (the training R^2 was just 0.4179), which was corrected by tuning C, epsilon and gamma, leading to a 133.64% increase in the training R^2 . The training results should thus not be interpreted as an indication of the quality of the model and should be compared to the test-set results in Section 4.6.2.

4.6.2 Test-set

Table 4.8 shows the improvement in MAE with Bayesian Optimization on the held-out test set. Every model improved, with SVR gaining the most as its MAE fell from 2.4382 to 1.7266 (29.19%), followed by CatBoost from 1.8436 to 1.5409 (16.42%), Random Forest from 1.7226 to 1.4670 (14.84%) and XGBoost from 2.0194 to 1.8841 (6.70%). This is shown graphically in Figure 4.18, with the optimized line being below the baseline at all points, indicating that tuning decreased the MAE in all four learners. The difference in the vertical lines shows the size of each gain, with the highest gain being

at SVR and the lowest at XGBoost. The shape of both lines is similar; they both increase at XGBoost and decrease at CatBoost, indicating that the relative ranking of algorithms is mostly maintained following tuning. A prominent exception is SVR which at base level is the weakest model, but at optimisation level, approaches the performance of Random Forest.

Table 4.7 MAE Improvement from Bayesian Optimization for Kuching
(Test Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	1.7226	1.4670	14.84
XGBoost	2.0194	1.8841	6.70
CatBoost	1.8429	1.5409	16.42
SVR	2.0316	1.7266	29.19

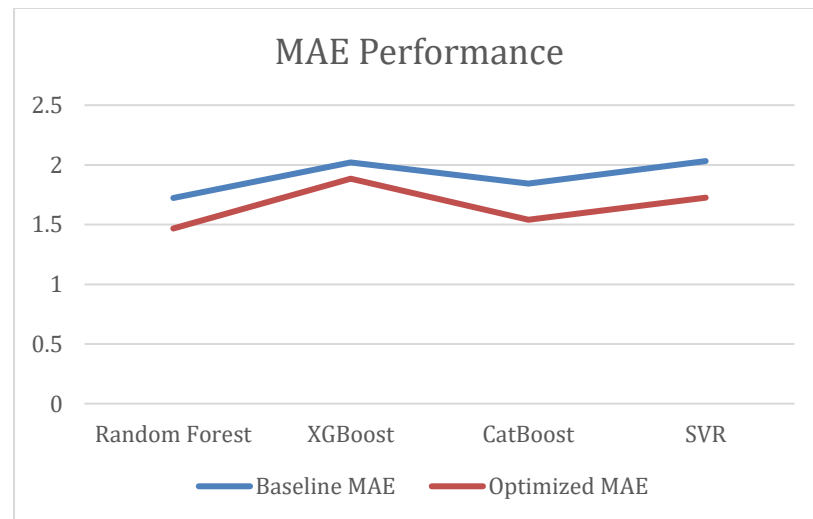


Figure 4.9 Comparison of baseline and optimized MAE across the four models (test set)

A similar comparison for RMSE is shown in Table 4.9. SVR again recorded the largest improvement, its RMSE falling from 4.2868 to 2.8100

(34.45%), followed by CatBoost from 2.8716 to 2.3555 (17.97%), Random Forest from 2.5323 to 2.2208 (12.30%) and XGBoost from 3.1470 to 2.8687 (8.84%). The results are shown in Figure 4.19, which once again shows that the optimized line is below the baseline for all models but with a much greater distance between the two at SVR when compared to the distance in MAE. The difference is more significant when considering RMSE, due to the fact that large errors are treated more harshly, suggesting that tuning has in fact helped to better capture SVR's performance for extreme PM2.5 values when haze occurs, which it had a poor performance modelling in pre-tuning. The optimized line is also visibly flatter than the baseline line and it is clear that the four models were able to achieve similar RMSE after tuning, whereas they spread apart widely at baseline.

Table 4.10 RMSE Improvement from Bayesian Optimization for Kuching (Test Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	2.5323	2.2208	12.30
XGBoost	3.1470	2.8687	8.84
CatBoost	2.8716	2.3555	17.97
SVR	4.2868	2.8100	34.45

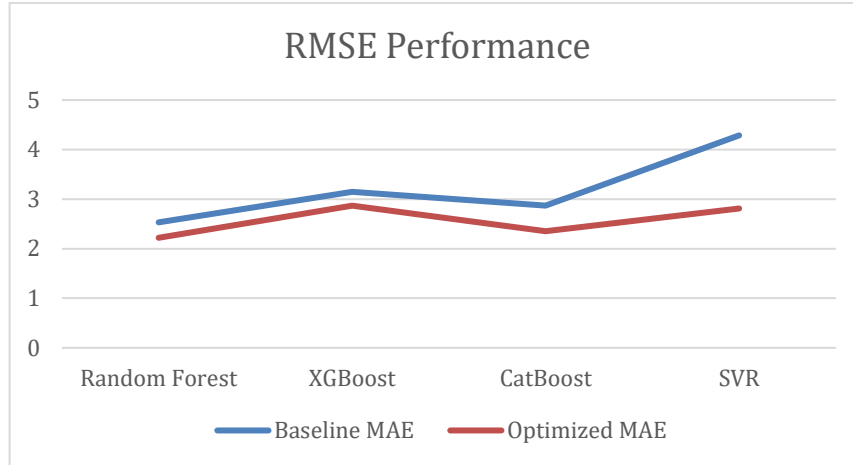


Figure 4.20 Comparison of baseline and optimized RMSE across the four models (test set)

The corresponding change in R^2 is reported in Table 4.11, with a higher value representing better performance (since the model is reverse engineered). SVR improved from 0.7790 to 0.9051 (16.19%), CatBoost from 0.9008 to 0.9333 (3.61%), XGBoost from 0.8809 to 0.9010 (2.28%) and Random Forest from 0.9229 to 0.9407 (1.93%). In each of the four models, the optimized line is above the baseline line as is depicted in figure 4.20. In percentage terms the gains seem smaller, due to the fact that R^2 is capped at 1.0, giving little room to models that already account for >90% of the variance. The most noticeable difference again is at SVR where the rise makes up for the significant drop in R^2 seen for the baseline line, and the tuned values of all four models lie within a relatively small band of values covering R^2 of 0.9010 to 0.9407, showing the convergence of R^2 observed during optimization.

Table 4.11 R^2 Improvement from Bayesian Optimization for Kuching
(Test Set)

Model	Baseline MAE	Optimized MAE	Improvement (%)
Random Forest	0.9229	0.9407	1.93
XGBoost	0.8809	0.9010	2.28
CatBoost	0.9008	0.9333	3.61
SVR	0.7790	0.9051	16.19

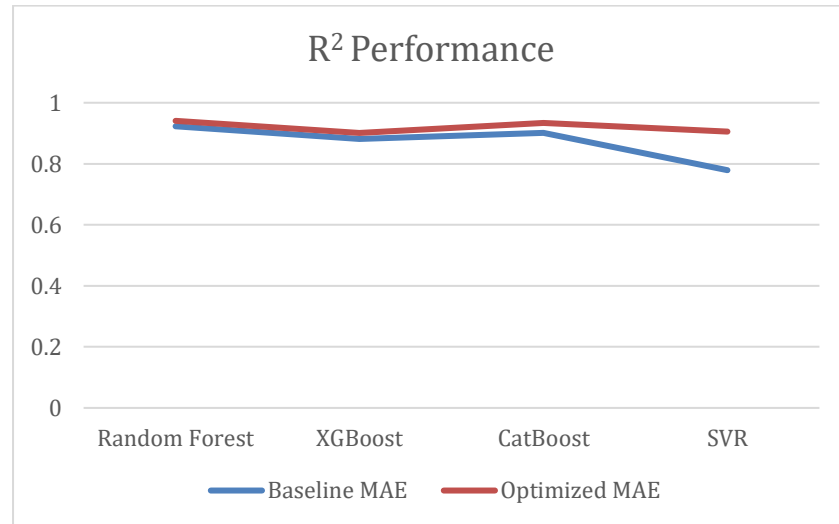


Figure 4.21 Comparison of baseline and optimized R^2 across the four models
(test set)

All these results together indicate that each model outperforms the others on all three metrics on the held-out test set, suggesting that Bayesian Optimization did not just overfit the training data. The improvement order is similar by all the three metrics, MAE, RMSE and R^2 , suggesting that the gains with tuning are not specific to any metric but are dependent on the algorithm's sensitivity to tuning it. The choice of C and γ parameters of the kernel methods had the largest impact on the performance of SVR, which is the weakest of the four learners with the default values, and was improved to perform competitively by optimizing the

parameters to $C = 100.0$, $\epsilon = 0.001$ and $\gamma = 0.1449$. The reason for Random Forest to have the least gain is that the bagging method already lowers variance, and the default parameters were nearer to optimal. The performance of the models was also similar, with the test R^2 range after tuning being much tighter, from 0.1439 at baseline to 0.0397 after tuning.

After optimization, Random Forest is the strongest individual learner on all three metrics (MAE 1.4670, RMSE 2.2208, R^2 0.9407), with CatBoost second (1.5409, 2.3555, 0.9333). These train–test MAE gaps illustrate the differences in generalization behaviour; XGBoost has the largest gap (0.9524), suggesting that this is an overfitting model, SVR has the smallest gap (0.1614), followed by Random Forest (0.2041) and CatBoost (0.2780). These diverse error profiles are just the kind of conditions where ensembles of classifiers can outperform their constituent classifiers, which is why the architecture proposed in Section 4.7 is desirable.

4.7 Stacking Ensemble

A stacking ensemble was built across the individually Bayesian-tuned models described in Section 3.6, rather than upon a single algorithm. The four tuned models (Random Forest, XGBoost, CatBoost and SVR) were used as base learners to train a separate XGBoost ($n_estimators = 200$, $learning_rate = 0.05$, $max_depth = 4$) model to learn how to best combine their outputs as a "meta-learner".

The base-learner predictions used to train the meta-learner are produced by 5-fold out-of-fold cross validation ($cv = 5$), in which the meta-learner never sees a prediction made by a base model on data that it was trained with. At this stage, the standard K-Fold cross validation was employed instead of TimeSeriesSplit as the latter does not guarantee that each sample gets exactly one out-of-fold prediction. Also, the original pollutant features were included with the base predictions in the

same manner so that the meta-learner could fine tune them directly from the raw features. The entire stack is displayed in the code segment in Figure 4.22.

```
estimators = [
    ('rf', best_rf),
    ('xgb', best_xgb),
    ('cat', best_cat),
    ('svr', best_svr),
]

stacking_model = StackingRegressor(
    estimators=estimators,
    final_estimator=XGBRegressor(
        n_estimators=200, learning_rate=0.05,
        max_depth=4, random_state=42, verbosity=0,
        objective='reg:squarederror'
    ),
    cv=5, # KFold(5) - compatible with cross_val_predict
    passthrough=True,
    n_jobs=-1,
)
```

Figure 4.22 Code snippet stacking ensemble

Table 4.12 Training-set performance of the tuned base learners and the stacking ensemble (Kuching, n = 2,045)

Model	RMSE	MAE	R ²
Random Forest (Optimized)	1.6959	1.2629	0.9870
XGBoost (Optimized)	1.2302	0.9317	0.9932
CatBoost (Optimized)	1.7108	1.2629	0.9868
SVR (Optimized)	2.2872	1.5652	0.9764
Stacking Ensemble	2.5691	1.3496	0.9702

Table 4.13 Test-set performance of the tuned base learners and the stacking ensemble (Kuching, n = 512)

Model	RMSE	MAE	R ²
Random Forest (Optimized)	2.2208	1.4670	0.9407
XGBoost (Optimized)	2.8687	1.8841	0.9010

Model	RMSE	MAE	R ²
CatBoost (Optimized)	2.3555	1.5409	0.9333
SVR (Optimized)	2.8100	1.7266	0.9051
Stacking Ensemble	2.4534	1.5983	0.9276

The results obtained by the stacking ensemble did not improve the best individual tuned model. Random Forest has the best R² (0.9407) and the lowest error (RMSE 2.2208, MAE 1.5983, for the Ensemble and 1.4670, for Random Forest) on the test set, followed by CatBoost (R² 0.9333) and the Ensemble (R² 0.9276). This can be attributed to the nature of the base learners: two of them (SVR, R² 0.9051 and XGBoost, R² 0.9010) perform significantly worse than Random Forest, but the meta-learner combines in predictions from the weaker models, which brings the overall prediction toward the center of the ensemble's predictions rather than above the strongest one.

For two reasons the ensemble is still kept. First, its test MAE of 1.5983 is significantly better than the average base learner: the MAE is a 20.32% lower than the mean baseline MAE of 2.0059, surpassing the target of 15% improvement in Objective 2, and 7.22% lower than the best baseline model. First, it significantly outperforms the average base learner: its test MAE of 1.5983 is 20.32% better than the mean baseline MAE of 2.0059, which exceeds the target of 15% improvement in Objective 2, and 7.22% lower than the best baseline model. Second, it is not tied to a single algorithm whose performance is best for the entire model life cycle, which is a useful measure of robustness when the model is used to forecast future data that may have features distinct from the current test set.

The train–test comparison indicates that the ensemble generalises well. It has a test training gap of 0.0426, and a gap of 0.2487 for MAE, both of which are not large, while Random Forest has a gap of 0.2041 (close), and much smaller than XGBoost (0.9524), which is a big gap, indicating overfitting remains. Interestingly, the training RMSE of the ensemble (2.5691) is larger than that of each of its base learners, suggesting that the meta-learner is not just memorizing

the training data, which is a desirable property for supporting the stability of the meta-learner on unseen observations.

4.8 AQI Forecast Accuracy

The PM2.5 predictions from the stacking ensemble were then converted to AQI values using the US EPA piecewise-linear breakpoint formula detailed in Chapter 3 and each prediction made in the AQI corresponding health category was assigned (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy and Hazardous). The AQI was calculated based on the predicted concentration of PM2.5 and not on maximum concentration of the pollutants in the sub index as is done in operational reports which are based on multiple pollutants.

In the 512 test observations, the predicted AQI category was the same as the actual category in 90.43% of the cases. The two lowest categories are dominant in the predicted distribution and include 355 records that are in the Good category and 143 in the Moderate category; 12 are in the Unhealthy for Sensitive Groups category and 2 are in the Hazardous category. This lower bound concentration is in line with the high right skew of PM2.5 seen in Table 4.14, with a majority of daily concentrations being low and elevated concentrations being present only sporadically. The few forecasts in the higher categories are due to occasional exceedance events of PM2.5 over the higher AQI thresholds, usually caused by the cross-border haze events that occur over Kuching. Table 4.14 shows the expected distribution of storm categories.

Table 4.14 Predicted AQI Category Distribution for Kuching Testing Set

Predicted AQI Category	Number of Records
Good	355
Moderate	143
Unhealthy for Sensitive Groups	12
Hazardous	2
AQI category match accuracy	90.43%

4.9 Prototype Web Application

The trained stacking ensemble was deployed as a lightweight web application which allows the prediction pipeline to be operated without any programming knowledge. The prototype uses a Flask backend to store the model artefacts and a browser-based front end to display the model artefacts to the user. The interface is geared towards a Department of Environment (DOE) policymaker audience, and not a technical machine-learning audience, so the underlying model details are put in the back seat. The application will take a dataset of monitoring stations, produce daily forecast predictions of PM_{2.5}, categorize the forecasts into the corresponding Air Quality Index (AQI) health levels, and return an interactive summary of the results.

4.9.1 System Architecture and Prediction Workflow

The prototype is based on a 3-layer architecture (presentation, application and model). The presentation layer provides a browser-based interface for user upload and display of the results. The application layer consists of a Flask server with the prediction endpoint, model-comparison endpoint, and the export endpoint and does the routing of the data from the interface to the model. A series of persisted artefacts are stored in the model layer, including the serialised stacking

ensemble (stacking_model.pkl), the fitted feature scaler (scaler.pkl), and the feature list that was retained during training (all_features.pkl), which are loaded at server start-up.

The prediction workflow starts with the user upload of a CSV or Excel file with the five monitoring variables (NO₂, O₃, CO, PM₁₀, and PM_{2.5}) in chronological order as shown in Figure 4.23. When the file is submitted, the server loads it and performs the same preprocessing and normalization as it did during training and then passes the features after the normalization to the stacking ensemble. The base learners (Random Forest, XGBoost, CatBoost, and SVR) make the first level predictions, and the second level prediction (the final PM_{2.5} estimate for each record) is made by combining the predictions of the base learners by the XGBoost meta-learner. The concentration for each prediction is then converted to an AQI value using the United States EPA piecewise-linear breakpoint formula and assigned to a health category. The results are displayed in the browser, and no data is stored in the browser so that the application can be used on any other station file.

4.9.2 User Interface and Visualization

The user interface includes an upload panel, a results dashboard, and two visualisations, as well as a detailed records table. These are described below with reference to the screen shot of the prototype. The upload panel will show the dataset and a reference guide showing the required input columns (NO₂, O₃, CO, PM₁₀ and PM_{2.5}) and the AQI category thresholds. This enables the user to verify that the data is expected to be in the desired format prior to conducting the analysis as illustrated in Figure 4.23.

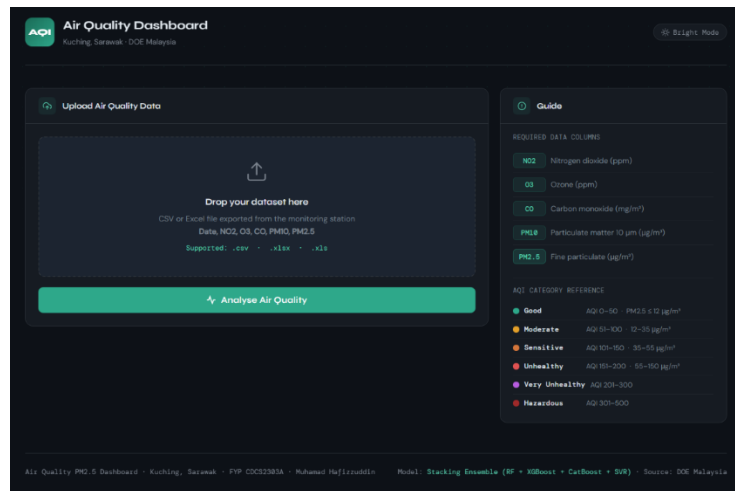


Figure 4.23 Prototype upload interface

After analysis, the results dashboard provides a summary of the forecast conditions in large, easy-to-read text. The headline includes the predicted predominant category, the number of readings processed, the percentage of readings that were classified as clean air, and a percentage that indicates the level of pollution alert (in the case of the model, 40%). A time-series graph below the summary displays the actual and ensemble-averaged PM2.5 concentrations and offers a direct visual comparison to assess the ability of the ensemble to capture the observed concentrations and locations of episodic high concentrations, as seen in Figure 4.24.



Figure 4.24 Prototype results dashboard

The distribution of predicted readings by AQI categories is summarised in a donut chart, which can be read easily and concisely, as seen in Figure 4.25.

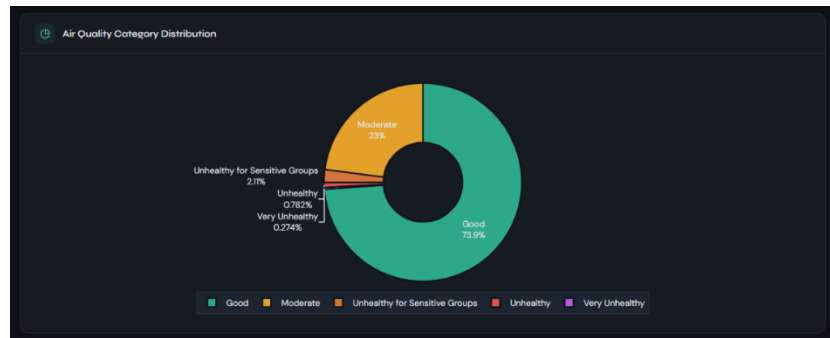


Figure 4.25 Result Donut chart

Finally, a records table contains the predicted AQI and the corresponding air-quality status for each reading, which would allow individual forecasts for each day to be checked, as illustrated in Figure 4.26. These components can combine the numerical output of the stacking ensemble into a user-friendly support decision map, appropriate for environmental monitoring and public reporting.

#	DATE / TIME	PREDICTED AQI	AIR QUALITY STATUS
1	#1	22	Good
2	#2	22	Good
3	#3	23	Good
4	#4	23	Good
5	#5	25	Good
6	#6	25	Good
7	#7	64	Moderate
8	#8	22	Good
9	#9	22	Good
10	#10	25	Good
11	#11	22	Good
12	#12	23	Good
13	#13	23	Good
14	#14	22	Good
15	#15	22	Good
16	#16	26	Good
17	#17	27	Good

Figure 4.26 Prediction results table (first 25 rows)

4.10 5-Fold Cross Validation

The experiment was also repeated with five different train/test split ratios (50/50, 60/40, 70/30, 80/20, 90/10) to ensure that the performance of the stacking ensemble was not the result of one specific split of the data. The split was always chronological (earliest records used for training; most recent for testing) in each of the runs. The temporal ordering of air quality observations is thus preserved and future information is not going to leak into training if the air quality observations were randomly shuffled.

The Bayesian-optimised hyperparameters used at each ratio of training-set size were those determined from Section 4.6 and were not re-tuned for this purpose. Min–Max scaler was re-trained for each run using just the training set for that run, and then applied to the test set for that run, which prevented leakage. The four optimized base learners (Random Forest, XGBoost, CatBoost and SVR) and the meta-learner (XGBoost) were re-trained on every

new training partition while the ensemble was tested on the held-out test partition with R^2 , RMSE and MAE.

Table 4.15 Stacking Ensemble Performance Across Five Train/Test Split Ratios

Split (Train/Test)	Train Size	Test Size	Train R^2	Test R^2	Test RMSE	Test MAE
50 / 50	1,278	1,279	0.9870	0.8812	2.4945	1.8724
60 / 40	1,534	1,023	0.9842	0.9118	2.2931	1.5839
70 / 30	1,789	768	0.9845	0.9131	2.4668	1.6561
80 / 20	2,045	512	0.9718	0.9263	2.4750	1.6074
90 / 10	2,301	256	0.9614	0.9516	1.3957	1.0642

The results show that the ensemble maintains consistently high predictive accuracy across all five partitions, with test R^2 ranging from 0.8812 at the 50/50 split to 0.9516 at the 90/10 split. As more training data is added, the R^2 of the test set continues to increase and the prediction error tends to decrease when more data is added to the training set, as the base learners are seeing a larger and more representative sample of the pollutant–PM_{2.5} relationship. The 80/20 split that has been used for the majority of this study does result in a test R^2 of 0.9263, a RMSE of 2.4750 and a MAE of 1.6074 which is very similar to the headline result reported in Section 4.7, and therefore represents a good and representative split rather than a coincidence.

The average test R^2 for the ensemble is displayed in the table 4.15 with a standard deviation of only 0.0255, making the ensemble very stable when tested across the five runs. This low variance shows a stable and good generalisation of the framework, and that its performance is not excessively sensitive to the actual size of the training partition used. All split ratios are well

above the objective of $R^2 > 0.80$, which was set for Objective 3, and the objective is satisfied in both noticeably higher and lower training proportions.

The 90/10 split yields the best individual performance (R^2 0.9516, RMSE 1.3957, MAE 1.0642), likely due in part to the larger training set, but also due to a smaller (256 records) but more recent testing window, where there was less extreme variability in PM2.5. The other extreme case is the 50/50 split implementation, which has half of the data used for training, and the biggest window in the test data of 1279 records; however, the ensemble still achieves a very good R^2 of 0.8812. Overall, this sensitivity analysis shows that the proposed Bayesian-optimised stacking framework offers robust results for PM2.5 predictions, regardless of the train/test configuration used.

4.11 Achievement of Research Objectives

The results from the experiments presented in this chapter were used to assess the extent to which the three research objectives outlined in Chapter 1 were achieved. Table 4.16 is a summary of the evidence that supports each objective and indicates the status of meeting that objective.

Table 4.16 Research Objective Achievement

No.	Research Objective	Evidence of Achievement	Status
1	To compare the predictive performance of advanced machine learning algorithms (Random Forest, XGBoost, CatBoost and SVR) in forecasting air quality.	All four algorithms were trained and evaluated on the Kuching dataset under both baseline and optimized configurations and compared using RMSE, MAE and R^2 . Random Forest was the strongest baseline learner across all three metrics (R^2 0.9229, RMSE 2.5323, MAE 1.7226) and remained the strongest after tuning (R^2 0.9407, RMSE	Achieved

No.	Research Objective	Evidence of Achievement	Status
		2.2208, MAE 1.4670), with CatBoost second (R^2 0.9333).	
2	To improve prediction performance using Bayesian Optimization for feature selection and hyperparameter tuning.	Bayesian Optimization was applied to both feature-subset selection and hyperparameter tuning for all four models. Every model improved on every metric, with SVR gaining the most (R^2 rising from 0.7790 to 0.9051, a 29.19% MAE reduction), followed by CatBoost (0.9008 to 0.9333, 16.42% MAE reduction) and Random Forest (0.9229 to 0.9407, 14.84% MAE reduction). No model degraded.	Achieved
3	To develop and validate a Stacking Ensemble integrating the optimized models, targeting $R^2 \geq 0.80$ and at least a 15% MAE reduction versus average baseline models.	The stacking ensemble achieved a test R^2 of 0.9276, comfortably exceeding the $R^2 \geq 0.80$ target, and reduced MAE by 20.32% relative to the average baseline (2.0059 to 1.5983), surpassing the 15% reduction target. Stability was further confirmed by the split sensitivity analysis, in which the ensemble exceeded $R^2 > 0.80$ across all five train/test ratios (mean R^2 0.9168 ± 0.0255).	Achieved

The three research goals were achieved. The relative performance of the four algorithms was evaluated through the comparative evaluation, and the evaluation result is that the performance of Bayesian Optimization algorithm is better than the other three algorithms on every item, and the stacking ensemble algorithm can reach the target quantitatively on the two items of Objective 3, and the R^2 value is 0.9276, and the MAE reduction rate is 20.32% compared with the average baseline. It should be mentioned that the ensemble did not outperform the best tuned individual tuned model (Random Forest— R^2 0.9407); however, Objective 3 was established in comparison to

the average baseline, and the ensemble outperformed this with a large margin as well as providing increased robustness against the reliance on a single model.

4.12 Discussion

The results are discussed with respect to the research goals and the broader body of literature on air quality prediction, base learners behaviour, role of Bayesian optimization and stacking, and practical implications and limitations of the study.

4.12.1 Comparative Performance of the Base Learners

Random Forest and the gradient-boosting learners performed the best, outperforming Support Vector Regression. Random Forest was the strongest model both at baseline (R^2 0.9229, RMSE 2.5323, MAE 1.7226) and after tuning (R^2 0.9407, RMSE 2.2208, MAE 1.4670), with CatBoost close behind (R^2 0.9333, MAE 1.5409). In terms of the R^2 , the weakest baseline learner was SVR (R^2 0.7790), which was sensitive to the heavy-tailed distribution of PM_{2.5} and the challenge of determining a universal kernel and regularization value for routine/daytime and haze/nighttime conditions. This difference in the base learner strengths is the rationale for an ensemble approach.

4.12.2 Effect of Bayesian Optimization and the Stacking Ensemble

The learners most sensitive to hyperparameter tuning (SVR (MAE reduced by 29.19%), CatBoost (16.42%)) saw the greatest improvements when hyperparameters were tuned using the Bayesian approach, whereas the others saw smaller improvements (Random

Forest (14.84%) and XGBoost (6.70%)). The test R^2 value of the stacking ensemble was 0.9276 and the MAE value was 1.5983, meeting both the criteria from Objective 3. It was, however, not able to outperform the best individual learner, the tuned Random Forest (R^2 0.9407). This implies that the composition of the ensemble: since two of the base learners (XGBoost and SVR) predict much lower than the RF, the meta-learner combines in lower predictions and moves towards the center of the ensemble range. However, the ensemble is robust because it is not relying on any single algorithm being the strongest, and is found to be stable across all the 5 split ratios (mean R^2 0.9168 ± 0.0255).

4.12.3 Practical Implications and Limitations

AQI category match accuracy of 90.43% means that the framework is well suited for public health advice as the most health-important categories are associated with higher levels of PM_{2.5}. The prototype web application illustrates how these predictions can be transferred to a non-technical user, from an offline modelling process to a decision support tool.

These findings have some limitations. The models have not included any meteorological parameters, including temperature, humidity and wind speed, which limits their ability to predict haze events and is the main source of error. Second, the numeric boundaries generate the categories for AQI, which means that values close to a boundary can be miscategorized even if the actual PM_{2.5} forecast is correct. Third, the study is limited to one region and timeframe, leaving the need for validation in other regions and further out in time. The limitations serve as a basis for future recommendations.

4.13 Summary

This chapter presented the experimental results of the proposed Bayesian Optimization–enhanced stacking ensemble framework for PM_{2.5} prediction in Kuching. The results of the main findings are as follows:

- 1) Bayesian Optimization identified PM₁₀ as the most prominent predictor (selection score 0.6833), followed by CO (0.2410); all four pollutant features were retained in the final model to preserve robustness and physical interpretability.
- 2) Bayesian hyperparameter tuning improved every model on every test metric without exception, with the largest gains for SVR and CatBoost and the smallest for XGBoost.
- 3) The stacking ensemble achieved a test R^2 of 0.9276, exceeding the $R^2 \geq 0.80$ target, and reduced MAE by 20.32% relative to the average baseline, surpassing the 15% target.
- 4) AQI category match accuracy reached 90.43%, indicating the framework is practically applicable for public air quality forecasting.

Prediction was found to be difficult in Kuching due to the highly variable PM_{2.5} concentration and the high concentration observed during a haze episode of 211.90 $\mu\text{g}/\text{m}^3$. Future work should include a consideration of meteorological characteristics and haze event indicators for better performance under these conditions.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

This chapter concludes the study. It revisits the research objectives set out in Chapter 1, summarises the main findings from the experimental results in Chapter 4, and discusses their implications for air quality forecasting in Kuching. It then outlines the limitations of the research and offers recommendations for future work, before closing with concluding remarks.

5.1 Introduction

This chapter reiterates the research objectives outlined in Chapter 1, highlights the key results obtained in Chapter 4 and discusses the implications of these results for air quality forecasting in Kuching. The objective of the study was to provide an enhanced system for daily prediction of PM_{2.5} concentration by implementing a stacking ensemble of machine learning models with Bayesian Optimization to accomplish both feature selection and hyperparameter optimization, and also map the predicted PM_{2.5} levels into Air Quality Index (AQI) health categories that can be utilized by public warnings. The chapter includes the main findings, the implications of these findings, the limitations of the study, future recommendations, and concluding remarks.

5.2 Summary of Main Findings

The study addressed three research objectives. Table 5.1 summarises the approach, key results and attainment status for each.

Table 5.1 Summary of Research Objectives and Findings

Objective	Approach	Key Results	Status
1. Feature Selection	Bayesian Optimization (Gaussian Process, Expected Improvement) over 5-fold TimeSeriesSplit to rank pollutant inputs.	PM10 ranked highest (0.6833) and was the only feature above the 0.5 threshold; CO (0.2410), O ₃ (0.2309) and NO ₂ (0.0160) fell below. Fallback retained all four inputs for robustness.	Achieved
2. Tuning & Ensemble	BO hyperparameter tuning of RF, XGBoost, CatBoost and SVR; combined via an XGBoost meta-learner with passthrough.	Largest gains for SVR: RMSE 4.2868 - 2.8100 (-34.45%), MAE 2.4382 - 1.7266 (-29.19%), R ² 0.7790 - 0.9051 (+16.19%). All models improved without exception.	Achieved
3. Evaluation & AQI	Evaluate against the R ² ≥ 0.80 benchmark and map predicted PM2.5 to AQI categories (US EPA breakpoints).	Stacking ensemble: R ² 0.9276, RMSE 2.4534, MAE 1.5983 µg/m ³ , exceeding the benchmark; 20.32% MAE reduction against average baseline. AQI category-match accuracy 90.43% across 512 test observations.	Achieved

5.2.1 First Objective

The first objective was to apply Bayesian Optimization to feature selection in order to identify the most relevant pollutant inputs for PM2.5 prediction. The Gaussian Process surrogate with the Expected Improvement acquisition function resulted in the highest selection score with PM10 (0.6833) being the only feature to pass the inclusion threshold (0.5), whereas CO (0.2410), O₃ (0.2309) and NO₂ (0.0160) did not. The large contribution of PM10 is due to its well known physical link with PM2.5, both containing the same emissions sources, such as transport, biomass burning and industry. Since less than 2 features

exceeded the threshold, all 4 pollutant inputs would be kept for training, which would maintain robust and physical interpretability. This goal was achieved: Bayesian Optimization was able to rank the inputs by predictive relevance, the ranking was used in the final feature configuration.

5.2.2 Second Objective

The second objective was to apply Bayesian Optimization to hyperparameter tuning and to construct a stacking ensemble for PM_{2.5} prediction. Five-fold out-of-fold predictions were used to tune and combine four base learners (Random Forest, XGBoost, CatBoost and SVR) with using a passthrough layer in XGBoost meta-learner. Tuning improved every model on every test metric, with the largest gains for the weakest baseline learner, SVR, whose RMSE fell from 4.2868 to 2.8100 $\mu\text{g}/\text{m}^3$ (-34.45%), MAE from 2.4382 to 1.7266 $\mu\text{g}/\text{m}^3$ (-29.19%), and R^2 rose from 0.7790 to 0.9051 (+16.19%).

The stacking ensemble had the best R^2 of 0.9276, RMSE of 2.4534 $\mu\text{g}/\text{m}^3$ and MAE of 1.5983 $\mu\text{g}/\text{m}^3$, which was a 20.32% decrease in MAE compared to the average baseline. It did not outperform the best individually tuned learner (Random Forest, R^2 0.9407), but it was able to bundle together four heterogeneous base learners into one, reducing the reliance on the performance of any individual algorithm, a stability that was supported by being tested across five train/test split ratios with a mean R^2 of 0.9168 ± 0.0255 . The aim was achieved.

5.2.3 Third Objective

The third objective was to evaluate the framework against the $R^2 \geq 0.80$ benchmark and to generate AQI categories suitable for public health advisories. The stacking ensemble obtained a test R^2 value of 0.9276,

which is much higher than the required level, as well as a significant decrease in the error value when compared with the base learners. The framework's ability to predict the PM_{2.5} in terms of AQI was found to have a 90.43% category-match accuracy for the 512 test observations, with most of the values being in the Good and Moderate category and a small number in the higher-risk bands. The goal was achieved: The framework meets the quantitative accuracy criterion and can be used for the reporting of air quality in the field of health.

5.3 Implications of Findings

The implications of the results to air quality modelling in tropical haze-prone areas like Sarawak are discussed. First, the prominence of PM₁₀ suggests that in the absence of meteorological input, the co-measured particulate fraction is the most useful predictor of PM_{2.5}, a practical consideration for monitoring agencies that lack many sensors due to the limited number of pollutants that can be measured to support reasonable estimates.

Second, the benefits of automated tuning are highly dependent on the base algorithm and the shape of the target distribution: for example, SVR provides the best benefits for data with heavy tails in the pollutant distribution, and least benefits for less heavy-tailed data.

Third, the ensemble was found to have an $R^2 \geq 0.80$, and it was also determined that it reduced MAE by 15%, as well as supporting the concept that diverse base learners offer complementary information to the meta-learner. Just as significant is the fact that the ensemble did not always perform best than each of the individually tuned learners, clearly reminding that stacking is robust but not necessarily superior. Finally, the prototype web application shows how these outputs can be made available to non-technical users like environmental officers and the public without relying on technical expertise, using a simple dashboard showing summary metrics, visualisations

and exportable reports, that turns the offline pipeline into a usable decision-support tool.

5.4 Limitations of the Research

While the objectives were met, several limitations should be considered when interpreting the findings. These define the scope of the results rather than undermining them.

1) Absence of meteorological features:

The models used pollutant concentrations only (NO₂, O₃, CO and PM₁₀), excluding temperature, humidity, wind speed and precipitation. This omission of weather and trans-boundary transport effects is the principal source of prediction error, particularly during episodic haze events.

2) Boundary misclassification in AQI mapping:

Because AQI categories are defined by numeric breakpoints, observations near a boundary may be misclassified even when the underlying PM_{2.5} forecast is accurate, imposing an inherent ceiling on category-match accuracy.

3) Sensitivity to optimization variability:

Bayesian Optimization is stochastic, so feature scores, tuned hyperparameters and final metrics may vary between runs. The reported figures derive from a specific pipeline execution and small run-to-run variation is expected.

4) Single-city, single-horizon scope:

The framework was developed and tested for Kuching only, at a daily forecast horizon. Its generalisability to other locations and temporal scales remains to be established.

5) Exclusion of lag-based temporal features:

The final framework used no lagged pollutant values, so it does not exploit the short- and long-term autocorrelation typically present in daily PM_{2.5} series and cannot anticipate the build-up and decay of haze across consecutive days.

5.5 Recommendations for Future Work

The limitations above point directly to several avenues for improving and extending the study. The following recommendations are intended to be specific, realistic, and connected to the findings of this work.

1) Integrate meteorological and event-based features:

Adding temperature, humidity, wind speed, rainfall and explicit haze-event indicators offers the most direct route to reducing error during episodic PM_{2.5} spikes.

2) Explore temporal modelling:

Reintroducing lagged pollutant features would capture the autocorrelation the current static model ignores. These could be combined with sequence models such as LSTM or temporal convolutional networks to represent pollutant momentum and anticipate haze build-up and decay.

3) Address boundary misclassification:

Probabilistic or distance-to-boundary weighting of the AQI mapping, or reporting prediction intervals alongside point estimates, would prevent borderline cases from being silently misclassified.

4) Improve robustness to optimization variability:

Running the optimization across multiple random seeds and reporting averaged metrics with confidence ranges would strengthen confidence in the results and aid reproducibility.

5) Extend to multiple cities and horizons:

Validating the framework for other Sarawak cities such as Bintulu and Miri, and across multi-day forecast horizons, would broaden its usefulness for regional air quality management.

6) Enhance the prototype for deployment:

Extending the web application with scheduled data ingestion, monitoring and alerting would allow it to operate continuously rather than on demand.

5.6 Conclusion

In this study, the performance of a stacking ensemble of machine learning models along with Bayesian Optimization for feature selection and hyperparameter tuning for the daily PM_{2.5} prediction system for Kuching was explored. The framework achieved test $R^2 = 0.9276$, an RMSE of $2.4534 \mu\text{g}/\text{m}^3$ and an MAE of $1.5983 \mu\text{g}/\text{m}^3$, with a 20.32% reduction in MAE compared to the average baseline, and a 90.43% accuracy in matching AQI categories suitable for public health advisories that exceeded the $R^2 \geq 0.80$ benchmark, demonstrating the three objectives were met: Bayesian Optimization successfully ranked the pollutant inputs by predictive relevance and informed the final feature configuration; the tuned base learners were combined into a stacking ensemble; and the resulting framework exceeded the $R^2 \geq 0.80$ benchmark with test $R^2 = 0.9276$, an RMSE of $2.4534 \mu\text{g}/\text{m}^3$ and an MAE of $1.5983 \mu\text{g}/\text{m}^3$, with 90.43% accuracy in matching AQI categories suitable for public health advisories.

In addition to the numerical results, the study provides a user-friendly and repeatable air quality prediction pipeline for tropical haze-prone regions, and a functioning prototype which allows air quality model results to be used by non-technical users. Openly conceding these limitations in scope and lack of meteorological input, it immediately serves as an agenda for further research on richer sets of features, modelling with time and a greater validation of the other features. Bringing the problem back to the one posed in Chapter 1, the

study shows that automated optimization and ensemble learning are feasible and effective as the basis for air quality forecasting in areas where they are struck by recurrent transboundary haze.

APPENDIX A

Ethics Approval Letter

BAHAGIAN PENYELIDIKAN & JARINGAN INDUSTRI RESEARCH & INDUSTRIAL NETWORK DIVISION	Universiti Teknologi MARA Cawangan Sarawak
UNIT PENGURUSAN PENYELIDIKAN RESEARCH MANAGEMENT UNIT (RMU)	Universiti Teknologi MARA Cawangan Sarawak Kampus Samarahan 2 94300 KOTA SAMARAHAN Tel: 082-678052/082-678061 Faks: 082-678065 e-mail: rmusarawak@uitm.edu.my
	 UNIVERSITI TEKNOLOGI MARA Cawangan Sarawak

Reference : 600-UITMKS (RMU.5/1/2) (99) JLD. 4
Our Reference: SWKBERC/U184/2026
Date : 1447H 19 ذو القعدة
6 Mei 2025

Azlina Binti Narawi
Muhamad Hafizzuddin Bin Haslin (2024693394)
Faculty of Computer and Mathematical
Sciences
Universiti Teknologi MARA Sarawak Branch
94300 Kota Samarahan
SARAWAK

Dear Azlina Binti Narawi,

APPROVAL LETTER – UiTM SARAWAK BRANCH ETHICS REVIEW COMMITTEE

Thank you for submitting your research proposal to UiTM Sarawak Branch Ethics Review Committee (BERC). After considering your application, the Committee approved your proposal titled "Improving Air Quality Prediction using Bayesian Optimization for Feature Selection and Hyperparameter Tuning" in Kuching, Sarawak.

Details of the approval are as follows:

Ref. number	SWKBERC/U184/2026
Approved Period	7 May 2026 - 31 August 2026
Authorized personnel	1. Azlina Binti Narawi 2. Muhamad Hafizzuddin Bin Haslin (2024693394)

The UiTM Sarawak Branch Ethics Review Committee operates in accordance to the-ICH Good Clinical Practice Guidelines, Malaysia Good Clinical Practice Guidelines and the Declaration of Helsinki. The approval of this project is conditional upon your continuing compliance with these guidelines and declaration.

If you require further information, please contact UiTM Sarawak BERC Secretariat at 082-678133 or email at rmusarawak@uitm.edu.my.

اوسها، تقوى، مليا

"MALAYSIA MADANI"
"BERKHIDMAT UNTUK NEGARA"

Yours sincerely,



Ts. DR. MOHAMAD IZWAN BIN ISMAIL
Chairman
UiTM Sarawak Branch Ethics Review Committee

APPENDIX B

Gant Chart

SEMESTER	SEMESTER 5														SEMESTER 6															
TASK	OCTOBER				NOVEMBER			DECEMBER			JANUARY				APRIL					MAY			JUNE				JULY			
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14		
Requirement Analysis																														
Preliminary Study																														
Data Collection																														
System Design																														
System Architecture and Workflow																														
Data Processing and Optimization Design																														
Machine Learning Model Design																														
Implementation																														
Data Preprocessing Implementation																														
Feature Selection (BO)																														
ML Model Implementation																														
Hyperparameter Tuning (BO)																														
Testing																														
Model Performance Evaluation																														
Model Comparison & Validation																														
Impact of Bayesian Optimization																														
Discussion of Results																														
Documentation																														
Writing Report																														

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